Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Tony Morris Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
6506 Vivian RD	Modesto	Stanislaus	95358

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X017-X063-X018-XXXX X022-X003-X001-XXXX

B. OPERATORS

Morris, Tony J. & Delfina R

Operator name: Morris, Tony J. & Delfina R

Telephone no.: (209) 000-0000 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 580596	Modesto	CA	95358

This operator is responsible for paying permit fees.

C. OWNERS

Morris, Tony J. & Delfina R

Legal owner name: Morris, Tony J. & Delfina R

Telephone no.: (209) 000-0000 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 580596	Modesto	CA	95358

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	420	460	400	420
Number under roof	1,700	350	0	0	0	0
Maximum number	1,700	350	550	560	440	420
Average number	1,700	350	420	460	400	420
Avg live weight (lbs)	1,400	1,450	950	700		

Predominant milk cow breed: Holstein

Average milk production: 60 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 58,234.08 *tons per reporting period*

Total nitrogen from manure: 733,349.35 *lbs per reporting period*

After ammonia losses (30% loss applied): 513,344.55 *lbs per reporting period*

Total phosphorus from manure: 117,253.22 *lbs per reporting period*

Total potassium from manure: 314,396.69 *lbs per reporting period*

Total salt from manure: 880,927.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 70,290,000 *gallons*

Total nitrogen generated: 210,317.53 *lbs*

Total phosphorus generated: 35,494.67 *lbs*

Total potassium generated: 382,846.99 *lbs*

Total salt generated: 3,306,457.57 *lbs*

	70,290,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	70,290,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
IW #1 Dairy	Ground water
IW #2 Carpenter	Ground water
IW #3 Vivian	Ground water
TID Lateral 2	Surface water
TID Lateral 2.5	Surface water

Source Description	Type
TID Lateral 3	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/20/2023	Solid commercial fertilizer *4-0-0*	37.80 *ton*	Dry-weight	0.1	4.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	3,020.98	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	3,020.98	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	Corral solids	1,080.00 *ton*	Dry-weight	49.9		9,450.00	2,220.00	11,300.00		57.30

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	10,226.41	2,402.40	12,228.41	620,077.68
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	10,226.41	2,402.40	12,228.41	620,077.68

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Christianson	32	32	2	manure	X017-X058-X052-XXXX X017-X058-X053-XXXX
Coelho 1	67	67	1	none	X022-X003-X001-XXXX
Coelho 2	35	35	2	process wastewater	X022-X003-X001-XXXX
Coelho 3	40	40	1	manure	X022-X003-X001-XXXX
Coito-Keyes	47	47	2	manure	X041-X042-X009-XXXX
Coito-Vivan	75	75	2	process wastewater	X022-X002-X051-XXXX X022-X002-X052-XXXX
Gomes	16	16	2	manure	X017-X058-X036-XXXX
Lacross	38	38	2	manure	X017-X058-X028-XXXX
M1	88	88	2	process wastewater	X022-X004-X014-XXXX X022-X004-X015-XXXX
M2	57	57	1	none	X022-X004-X013-XXXX
Machado	57	57	2	process wastewater	X022-X002-X027-XXXX
Morris 1	3	3	1	process wastewater	X017-X063-X018-XXXX
Morris 2	12	12	1	process wastewater	X017-X063-X018-XXXX
Soares	77	77	1	none	X022-X003-X014-XXXX
Sousa	56	56	2	manure	X017-X058-X044-XXXX X017-X058-X045-XXXX
Sousa Carpenter	90	90	1	process wastewater	X017-X063-X008-XXXX
Sung-Ken	60	60	2	process wastewater	X017-X063-X010-XXXX
Van Rys 1	63	63	3	process wastewater	X017-X063-X004-XXXX X017-X063-X005-XXXX
Van Rys 2	36	36	1	process wastewater	X017-X063-X004-XXXX X017-X063-X005-XXXX
Totals for areas that were used for application	748	748	28		
Totals for areas that were not used for application	201	201	3		
Land application area totals	949	949	31		

B. CROPS AND HARVESTS

Christianson

Field name: Christianson

Christianson

11/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 32 Plant date: 11/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	487.00 *ton*	As-is		73.0	6,160.00	1,730.00	16,200.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.22	187.50	52.66	493.09	977.96

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 32 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	871.00 *ton*	As-is		69.0	3,660.00	880.00	4,630.00		4.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.22	199.24	47.91	252.05	826.91

Coelho 1

Field name: Coelho 1

11/18/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 67 Plant date: 11/18/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	412.00 *ton*	As-is		10.0	30,000.00	2,700.00	21,000.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	6.15	368.96	33.21	258.27	0.00

Coelho 2

Field name: Coelho 2

11/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	388.00 *ton*	As-is		71.6	6,990.00	1,530.00	15,600.00		13.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	11.09	154.98	33.92	345.87	824.87

06/25/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 35 Plant date: 06/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/02/2023	357.00 *ton*	As-is		62.4	6,940.00	1,390.00	12,400.00		15.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	11.00	121.00	18.70	132.00	0.00
Total actual harvest content	10.20	141.58	28.36	252.96	1,173.57

Coelho 3

Field name: Coelho 3

11/10/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 40 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	288.00 *ton*	As-is		10.0	30,000.00	2,700.00	21,000.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	7.20	432.00	38.88	302.40	0.00

Coito-Keyes

Field name: Coito-Keyes

11/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 47 Plant date: 11/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	712.00 *ton*	As-is		70.4	6,510.00	1,690.00	16,600.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.15	197.24	51.20	502.94	1,022.37

06/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 47 Plant date: 06/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	1,456.00 *ton*	As-is		71.0	3,600.00	870.00	4,340.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	30.98	223.05	53.90	268.90	1,006.19

Coito-Vivan

Field name: Coito-Vivan

11/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 75 Plant date: 11/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	1,089.00 *ton*	As-is		72.2	6,280.00	1,670.00	15,300.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.52	182.37	48.50	444.31	960.70

Coito-Vivan

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/14/2023	2,303.00 *ton*	As-is		69.2	3,510.00	1,200.00	5,500.00		4.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	30.71	215.56	73.70	337.77	907.93

Gomes

Field name: Gomes

11/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 16 Plant date: 11/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	289.00 *ton*	As-is		74.4	6,840.00	1,590.00	14,300.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	18.06	247.10	57.44	516.59	1,146.75

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	489.00 *ton*	As-is		70.0	3,480.00	950.00	4,810.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	30.56	212.72	58.07	294.01	990.23

Lacross

Field name: Lacross

11/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 11/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	588.00 *ton*	As-is		73.4	5,850.00	1,540.00	14,600.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.47	181.04	47.66	451.83	938.45

06/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,103.00 *ton*	As-is		70.4	3,290.00	880.00	4,950.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	29.03	190.99	51.09	287.36	893.55

M1

Field name: M1

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 88 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/06/2023	1,385.00 *ton*	As-is		71.8	6,510.00	1,690.00	17,200.00		12.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.74	204.92	53.20	541.41	1,127.33

M1

06/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 88 Plant date: 06/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	2,451.00 *ton*	As-is		69.0	3,630.00	890.00	4,950.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.85	202.21	49.58	275.74	1,001.57

M2

Field name: M2

11/22/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 57 Plant date: 11/22/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	386.00 *ton*	As-is		10.0	30,000.00	2,700.00	21,000.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	6.77	406.32	36.57	284.42	0.00

Machado

Field name: Machado

Machado

11/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 57 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	962.00 *ton*	As-is		73.6	6,200.00	1,690.00	16,100.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	16.88	209.28	57.04	543.45	1,096.07

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 57 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	1,903.00 *ton*	As-is		69.2	3,390.00	890.00	4,900.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	33.39	226.36	59.43	327.18	1,069.42

Morris 1

Field name: Morris 1

11/28/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 3 Plant date: 11/28/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	20.00 *ton*	As-is		10.0	30,000.00	2,700.00	21,000.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	6.67	400.00	36.00	280.00	0.00

Morris 2

Field name: Morris 2

11/28/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 12 Plant date: 11/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	84.00 *ton*	As-is		10.0	30,000.00	2,700.00	21,000.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	7.00	420.00	37.80	294.00	0.00

Soares

Field name: Soares

11/23/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 77 Plant date: 11/23/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	506.00 *ton*	As-is		10.0	30,000.00	2,700.00	21,000.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	6.57	394.29	35.49	276.00	0.00

Sousa

Field name: Sousa

Sousa

11/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 56 Plant date: 11/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	921.00 *ton*	As-is		71.4	6,610.00	1,660.00	16,900.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	16.45	217.42	54.60	555.89	1,119.48

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 56 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	1,702.00 *ton*	As-is		70.8	3,210.00	830.00	4,080.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	30.39	195.12	50.45	248.01	993.97

Sousa Carpenter

Field name: Sousa Carpenter

12/08/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 90 Plant date: 12/08/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	531.00 *ton*	As-is		10.0	30,000.00	2,700.00	21,000.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	5.90	354.00	31.86	247.80	0.00

Sung-Ken

Field name: Sung-Ken

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 60 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	846.00 *ton*	As-is		71.6	7,360.00	1,680.00	18,700.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.10	207.55	47.38	527.34	904.99

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 60 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	1,829.00 *ton*	As-is		71.2	3,510.00	780.00	4,540.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	30.48	213.99	47.55	276.79	1,071.06

Van Rys 1

Field name: Van Rys 1

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 63 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	781.00 *ton*	As-is		67.0	7,760.00	1,850.00	20,100.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	12.40	192.40	45.87	498.35	900.01

Van Rys 1

05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 63 Plant date: 05/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/30/2023	1,894.00 *ton*	As-is		68.0	3,940.00	920.00	5,260.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	30.06	236.90	55.32	316.27	1,192.92

09/02/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 63 Plant date: 09/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/02/2023	412.00 *ton*	As-is		67.4	6,200.00	1,010.00	10,200.00		20.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	6.54	81.09	13.21	133.41	869.83

Van Rys 2

Field name: Van Rys 2

11/28/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 36 Plant date: 11/28/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	224.00 *ton*	As-is		10.0	30,000.00	2,700.00	21,000.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	6.22	373.33	33.60	261.33	0.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

Christianson - 11/14/2022: Oats, silage-soft dough

Field name: Christianson

Crop: Oats, silage-soft dough — Plant date: 11/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator	Separator solids	219.04	48.49	102.91	2,328.19	1,030.00 *ton*
Application event totals		219.04	48.49	102.91	2,328.19	

10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	0.00	5,265,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	22.81	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.81	

Christianson - 06/20/2023: Corn, silage

Field name: Christianson

Crop: Corn, silage — Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	245.60	57.70	293.68	14,891.91	830.00 *ton*
Application event totals		245.60	57.70	293.68	14,891.91	

Christianson - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	28.52	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	24.71	5,265,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	22.81	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	22.81	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	23.76	5,062,500.00 *gal*
Application event totals		0.00	0.00	0.00	23.76	

Christianson - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	22.81	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	20.91	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	20.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	20.91	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	20.91	

Coelho 1 - 11/18/2021: Alfalfa, hay

Field name: Coelho 1

Crop: Alfalfa, hay

Plant date: 11/18/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	35.41	10,530,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	34.05	10,125,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.05	

Coelho 1 - 11/18/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	34.05	10,125,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.05	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	34.05	10,125,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.05	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	32.69	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.69	

08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	32.69	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.69	

09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	31.33	9,315,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.33	

10/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	29.96	8,910,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.96	

Coelho 2 - 11/11/2022: Oats, silage-soft dough

Coelho 2 - 11/11/2022: Oats, silage-soft dough

Field name: Coelho 2

Crop: Oats, silage-soft dough

Plant date: 11/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	117.81	20.94	185.19	1,407.68	900,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	0.00	6,075,000.00 *gal*
Application event totals		117.81	20.94	185.19	1,407.68	

04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.45	15.93	105.17	609.85	870,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	37.80	5,872,500.00 *gal*
Application event totals		68.45	15.93	105.17	647.66	

Coelho 2 - 06/25/2023: Sudangrass, silage

Field name: Coelho 2

Crop: Sudangrass, silage

Plant date: 06/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.09	15.38	101.54	588.82	840,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	36.50	5,670,000.00 *gal*
Application event totals		66.09	15.38	101.54	625.32	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	36.50	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.50	

Coelho 2 - 06/25/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.00	7.05	131.30	1,383.65	780,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	33.89	5,265,000.00 *gal*
Application event totals		61.00	7.05	131.30	1,417.54	

09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.00	7.05	131.30	1,383.65	780,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	33.89	5,265,000.00 *gal*
Application event totals		61.00	7.05	131.30	1,417.54	

Coelho 3 - 11/10/2022: Alfalfa, hay

Field name: Coelho 3

Crop: Alfalfa, hay

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	310.94	180.69	1,004.57	12,089.66	720.00 *ton*
Application event totals		310.94	180.69	1,004.57	12,089.66	

10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Coelho 3 - 11/10/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	36.50	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	34.22	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	34.22	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	34.22	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	35.36	6,277,500.00 *gal*
Application event totals		0.00	0.00	0.00	35.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	33.08	5,872,500.00 *gal*
Application event totals		0.00	0.00	0.00	33.08	

Coelho 3 - 11/10/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	31.94	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.94	

10/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	31.94	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.94	

Coito-Keyes - 11/12/2022: Oats, silage-soft dough

Field name: Coito-Keyes

Crop: Oats, silage-soft dough

Plant date: 11/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator	Separator solids	220.08	48.72	103.40	2,339.25	1,520.00 *ton*
Application event totals		220.08	48.72	103.40	2,339.25	

10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	0.00	7,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	26.17	7,020,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.17	

Coito-Keyes - 06/25/2023: Corn, silage

Field name: Coito-Keyes

Crop: Corn, silage

Plant date: 06/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator	Separator solids	249.07	34.49	89.41	1,628.53	1,400.00 *ton*
Application event totals		249.07	34.49	89.41	1,628.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	32.22	8,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	28.19	7,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	29.20	7,830,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	28.19	7,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.19	

Coito-Keyes - 06/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	26.17	7,020,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	24.16	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	24.16	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	24.16	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.16	

Coito-Vivan - 11/13/2022: Oats, silage-soft dough

Field name: Coito-Vivan

Crop: Oats, silage-soft dough

Plant date: 11/13/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	117.28	20.85	184.36	1,401.43	1,920,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	0.00	12,960,000.00 *gal*
Application event totals		117.28	20.85	184.36	1,401.43	

Coito-Vivan - 11/13/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.25	8.20	54.16	314.04	960,000.00 *gal*
Application event totals		35.25	8.20	54.16	314.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.50	16.41	108.31	628.08	1,920,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	38.93	12,960,000.00 *gal*
Application event totals		70.50	16.41	108.31	667.01	

Coito-Vivan - 06/17/2023: Corn, silage

Field name: Coito-Vivan

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.08	12.82	84.62	490.69	1,500,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	40.56	13,500,000.00 *gal*
Application event totals		55.08	12.82	84.62	531.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	38.93	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	38.93	

Coito-Vivan - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.36	5.82	108.40	1,142.40	1,380,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	37.31	12,420,000.00 *gal*
Application event totals		50.36	5.82	108.40	1,179.71	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.17	5.57	103.69	1,092.73	1,320,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	35.69	11,880,000.00 *gal*
Application event totals		48.17	5.57	103.69	1,128.42	

08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	45.98	5.31	98.98	1,043.06	1,260,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	34.07	11,340,000.00 *gal*
Application event totals		45.98	5.31	98.98	1,077.13	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.17	5.57	103.69	1,092.73	1,320,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	35.69	11,880,000.00 *gal*
Application event totals		48.17	5.57	103.69	1,128.42	

09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.79	5.06	94.27	993.39	1,200,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	32.45	10,800,000.00 *gal*
Application event totals		43.79	5.06	94.27	1,025.83	

Coito-Vivan - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	32.45	10,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.45	

Gomes - 11/14/2022: Oats, silage-soft dough

Field name: Gomes

Crop: Oats, silage-soft dough

Plant date: 11/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	226.73	131.76	732.50	8,815.38	210.00 *ton*
Application event totals		226.73	131.76	732.50	8,815.38	

10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	0.00	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	22.81	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.81	

Gomes - 06/20/2023: Corn, silage

Field name: Gomes

Crop: Corn, silage

Plant date: 06/20/2023

Gomes - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	248.56	58.39	297.22	15,071.33	420.00 *ton*
Application event totals		248.56	58.39	297.22	15,071.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	30.42	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	30.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	28.52	3,037,500.00 *gal*
Application event totals		0.00	0.00	0.00	28.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	26.62	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	28.52	3,037,500.00 *gal*
Application event totals		0.00	0.00	0.00	28.52	

Gomes - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	26.62	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	26.62	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	22.81	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	19.01	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	19.01	

Lacross - 11/14/2022: Oats, silage-soft dough

Field name: Lacross

Crop: Oats, silage-soft dough

Plant date: 11/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator	Separator solids	214.90	47.58	100.96	2,284.17	1,200.00 *ton*
Application event totals		214.90	47.58	100.96	2,284.17	

Lacross - 11/14/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	28.02	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.02	

Lacross - 06/21/2023: Corn, silage

Field name: Lacross

Crop: Corn, silage

Plant date: 06/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	249.18	58.54	297.96	15,109.11	1,000.00 *ton*
Application event totals		249.18	58.54	297.96	15,109.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	31.75	6,885,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

Lacross - 06/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	28.02	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	28.95	6,277,500.00 *gal*
Application event totals		0.00	0.00	0.00	28.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	28.02	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	26.15	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	22.41	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	24.28	5,265,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.28	

Lacross - 06/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	22.41	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.41	

M1 - 11/15/2022: Oats, silage-soft dough

Field name: M1

Crop: Oats, silage-soft dough Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	124.95	22.21	196.41	1,493.00	2,400,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	0.00	16,200,000.00 *gal*
Application event totals		124.95	22.21	196.41	1,493.00	

04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.59	15.73	103.85	602.21	2,160,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	37.33	14,580,000.00 *gal*
Application event totals		67.59	15.73	103.85	639.54	

M1 - 06/27/2023: Corn, silage

Field name: M1

Crop: Corn, silage Plant date: 06/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

M1 - 06/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.08	13.98	92.31	535.29	1,920,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	33.18	12,960,000.00 *gal*
Application event totals		60.08	13.98	92.31	568.48	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	31.11	12,150,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.11	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.99	6.47	120.51	1,269.96	1,800,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	31.11	12,150,000.00 *gal*
Application event totals		55.99	6.47	120.51	1,301.07	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.12	6.25	116.49	1,227.63	1,740,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	30.07	11,745,000.00 *gal*
Application event totals		54.12	6.25	116.49	1,257.70	

09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.25	6.04	112.48	1,185.29	1,680,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	29.03	11,340,000.00 *gal*
Application event totals		52.25	6.04	112.48	1,214.33	

M1 - 06/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.25	6.04	112.48	1,185.29	1,680,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	29.03	11,340,000.00 *gal*
Application event totals		52.25	6.04	112.48	1,214.33	

09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	26.96	10,530,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.96	

10/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	26.96	10,530,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.96	

M2 - 11/22/2018: Alfalfa, hay

Field name: M2

Crop: Alfalfa, hay

Plant date: 11/22/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	41.62	10,530,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.62	

04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	40.02	10,125,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.02	

M2 - 11/22/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	38.42	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	38.42	

06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	38.42	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	38.42	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	38.42	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	38.42	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	36.82	9,315,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.82	

09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	35.22	8,910,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.22	

09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	35.22	8,910,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.22	

Machado - 11/16/2022: Oats, silage-soft dough

Machado - 11/16/2022: Oats, silage-soft dough

Field name: Machado

Crop: Oats, silage-soft dough

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	115.74	20.58	181.94	1,382.99	1,440,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	0.00	9,720,000.00 *gal*
Application event totals		115.74	20.58	181.94	1,382.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.18	12.14	80.16	464.86	1,080,000.00 *gal*
Application event totals		52.18	12.14	80.16	464.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.98	13.49	89.07	516.51	1,200,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	32.02	8,100,000.00 *gal*
Application event totals		57.98	13.49	89.07	548.53	

Machado - 06/16/2023: Corn, silage

Field name: Machado

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.87	14.17	93.53	542.34	1,260,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	33.62	8,505,000.00 *gal*
Application event totals		60.87	14.17	93.53	575.96	

Machado - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	32.02	8,100,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.98	5.66	105.43	1,111.03	1,020,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	27.22	6,885,000.00 *gal*
Application event totals		48.98	5.66	105.43	1,138.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.86	5.99	111.63	1,176.38	1,080,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	28.82	7,290,000.00 *gal*
Application event totals		51.86	5.99	111.63	1,205.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.98	5.66	105.43	1,111.03	1,020,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	27.22	6,885,000.00 *gal*
Application event totals		48.98	5.66	105.43	1,138.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.10	5.33	99.23	1,045.67	960,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	25.61	6,480,000.00 *gal*
Application event totals		46.10	5.33	99.23	1,071.29	

Machado - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.10	5.33	99.23	1,045.67	960,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	25.61	6,480,000.00 *gal*
Application event totals		46.10	5.33	99.23	1,071.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	26.42	6,682,500.00 *gal*
Application event totals		0.00	0.00	0.00	26.42	

Morris 1 - 11/28/2020: Alfalfa, hay

Field name: Morris 1

Crop: Alfalfa, hay

Plant date: 11/28/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.08	12.82	84.62	490.69	60,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	31.54	540,000.00 *gal*
Application event totals		55.08	12.82	84.62	522.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	31.54	540,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.54	

Morris 1 - 11/28/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	31.54	540,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	31.54	540,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	31.54	540,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	23.66	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	23.66	405,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.74	6.33	117.83	1,241.74	60,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	23.66	405,000.00 *gal*
Application event totals		54.74	6.33	117.83	1,265.39	

Morris 2 - 11/28/2022: Alfalfa, hay

Field name: Morris 2

Crop: Alfalfa, hay

Plant date: 11/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.85	16.02	105.77	613.36	300,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	29.57	2,025,000.00 *gal*
Application event totals		68.85	16.02	105.77	642.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	29.57	2,025,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	26.62	1,822,500.00 *gal*
Application event totals		0.00	0.00	0.00	26.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	26.62	1,822,500.00 *gal*
Application event totals		0.00	0.00	0.00	26.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	23.66	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.66	

Morris 2 - 11/28/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	23.66	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	23.66	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.74	6.33	117.83	1,241.74	240,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	23.66	1,620,000.00 *gal*
Application event totals		54.74	6.33	117.83	1,265.39	

Soares - 11/23/2018: Alfalfa, hay

Field name: Soares

Crop: Alfalfa, hay

Plant date: 11/23/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	39.11	13,365,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	36.74	12,555,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.74	

Soares - 11/23/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	37.92	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	35.55	12,150,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	36.74	12,555,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	35.55	12,150,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	34.37	11,745,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	34.37	11,745,000.00 *gal*
Application event totals		0.00	0.00	0.00	34.37	

Sousa - 11/14/2022: Oats, silage-soft dough

Sousa - 11/14/2022: Oats, silage-soft dough

Field name: Sousa

Crop: Oats, silage-soft dough

Plant date: 11/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	231.35	134.44	747.45	8,995.29	750.00 *ton*
Application event totals		231.35	134.44	747.45	8,995.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	0.00	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	26.07	9,720,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.07	

Sousa - 06/20/2023: Corn, silage

Field name: Sousa

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	245.18	57.60	293.17	14,866.28	1,450.00 *ton*
Application event totals		245.18	57.60	293.17	14,866.28	

Sousa - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	27.16	10,125,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	16.00	0.00	0.00	0.00	
Application event totals		16.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	23.90	8,910,000.00 *gal*
Application event totals		0.00	0.00	0.00	23.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	24.99	9,315,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	22.81	8,505,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	23.36	8,707,500.00 *gal*
Application event totals		0.00	0.00	0.00	23.36	

Sousa - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	21.73	8,100,000.00 *gal*
Application event totals		0.00	0.00	0.00	21.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	22.81	8,505,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2	Surface water	0.00	0.00	0.00	20.64	7,695,000.00 *gal*
Application event totals		0.00	0.00	0.00	20.64	

Sousa Carpenter - 12/08/2019: Alfalfa, hay

Field name: Sousa Carpenter

Crop: Alfalfa, hay

Plant date: 12/08/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.09	15.38	101.54	588.82	2,160,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	28.39	14,580,000.00 *gal*
Application event totals		66.09	15.38	101.54	617.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	27.60	14,175,000.00 *gal*
Application event totals		0.00	0.00	0.00	27.60	

Sousa Carpenter - 12/08/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	26.81	13,770,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	26.81	13,770,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	25.24	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	25.24	12,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	24.45	12,555,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.74	6.33	117.83	1,241.74	1,800,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	23.66	12,150,000.00 *gal*
Application event totals		54.74	6.33	117.83	1,265.39	

Sung-Ken - 11/17/2022: Oats, silage-soft dough

Field name: Sung-Ken

Crop: Oats, silage-soft dough

Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	105.37	18.73	165.64	1,259.09	1,380,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	0.00	9,315,000.00 *gal*
Application event totals		105.37	18.73	165.64	1,259.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	99.14	23.07	152.31	883.23	2,160,000.00 *gal*
IW #2 Carpenter	Ground water	9.65	0.00	0.00	201.88	2,268,000.00 *gal*
Application event totals		108.79	23.07	152.31	1,085.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.83	13.46	88.85	515.22	1,260,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	24.84	8,505,000.00 *gal*
Application event totals		57.83	13.46	88.85	540.06	

Sung-Ken - 06/15/2023: Corn, silage

Field name: Sung-Ken

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.08	12.82	84.62	490.69	1,200,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	23.66	8,100,000.00 *gal*
Application event totals		55.08	12.82	84.62	514.34	

Sung-Ken - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	22.48	7,695,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.27	5.69	106.05	1,117.56	1,080,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	21.29	7,290,000.00 *gal*
Application event totals		49.27	5.69	106.05	1,138.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.53	5.38	100.16	1,055.48	1,020,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	20.11	6,885,000.00 *gal*
Application event totals		46.53	5.38	100.16	1,075.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.27	5.69	106.05	1,117.56	1,080,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	21.29	7,290,000.00 *gal*
Application event totals		49.27	5.69	106.05	1,138.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.53	5.38	100.16	1,055.48	1,020,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	20.11	6,885,000.00 *gal*
Application event totals		46.53	5.38	100.16	1,075.58	

Sung-Ken - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.79	5.06	94.27	993.39	960,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	18.93	6,480,000.00 *gal*
Application event totals		43.79	5.06	94.27	1,012.32	

09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	18.93	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	18.93	

Van Rys 1 - 11/17/2022: Oats, silage-soft dough

Field name: Van Rys 1

Crop: Oats, silage-soft dough Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	104.72	18.62	164.61	1,251.27	1,440,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	0.00	11,016,000.00 *gal*
Application event totals		104.72	18.62	164.61	1,251.27	

01/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.68	18.31	120.88	700.98	1,800,000.00 *gal*
IW #1 Dairy	Ground water	8.16	0.00	0.00	178.34	1,980,000.00 *gal*
Application event totals		86.84	18.31	120.88	879.32	

Van Rys 1 - 11/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.70	13.43	88.65	514.05	1,320,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	28.09	10,098,000.00 *gal*
Application event totals		57.70	13.43	88.65	542.14	

Van Rys 1 - 05/16/2023: Corn, silage

Field name: Van Rys 1

Crop: Corn, silage

Plant date: 05/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.70	13.43	88.65	514.05	1,320,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	28.09	10,098,000.00 *gal*
Application event totals		57.70	13.43	88.65	542.14	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	25.54	9,180,000.00 *gal*
Application event totals		0.00	0.00	0.00	25.54	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.14	6.02	112.22	1,182.61	1,200,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	25.54	9,180,000.00 *gal*
Application event totals		52.14	6.02	112.22	1,208.14	

Van Rys 1 - 05/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.53	5.72	106.61	1,123.48	1,140,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	24.26	8,721,000.00 *gal*
Application event totals		49.53	5.72	106.61	1,147.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.92	5.42	101.00	1,064.35	1,080,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	22.98	8,262,000.00 *gal*
Application event totals		46.92	5.42	101.00	1,087.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.92	5.42	101.00	1,064.35	1,080,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	22.98	8,262,000.00 *gal*
Application event totals		46.92	5.42	101.00	1,087.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.32	5.12	95.39	1,005.21	1,020,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	21.71	7,803,000.00 *gal*
Application event totals		44.32	5.12	95.39	1,026.92	

Van Rys 1 - 09/02/2023: Sudangrass, silage

Field name: Van Rys 1

Crop: Sudangrass, silage

Plant date: 09/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Van Rys 1 - 09/02/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	22.98	8,262,000.00 *gal*
Application event totals		0.00	0.00	0.00	22.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.32	5.12	95.39	1,005.21	1,020,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	21.71	7,803,000.00 *gal*
Application event totals		44.32	5.12	95.39	1,026.92	

Van Rys 2 - 11/28/2020: Alfalfa, hay

Field name: Van Rys 2

Crop: Alfalfa, hay

Plant date: 11/28/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.03	18.16	119.88	695.14	1,020,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	33.52	6,885,000.00 *gal*
Application event totals		78.03	18.16	119.88	728.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	31.54	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.54	

Van Rys 2 - 11/28/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	31.54	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	33.52	6,885,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	31.28	6,426,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	26.81	5,508,000.00 *gal*
Application event totals		0.00	0.00	0.00	26.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 2.5	Surface water	0.00	0.00	0.00	29.05	5,967,000.00 *gal*
Application event totals		0.00	0.00	0.00	29.05	

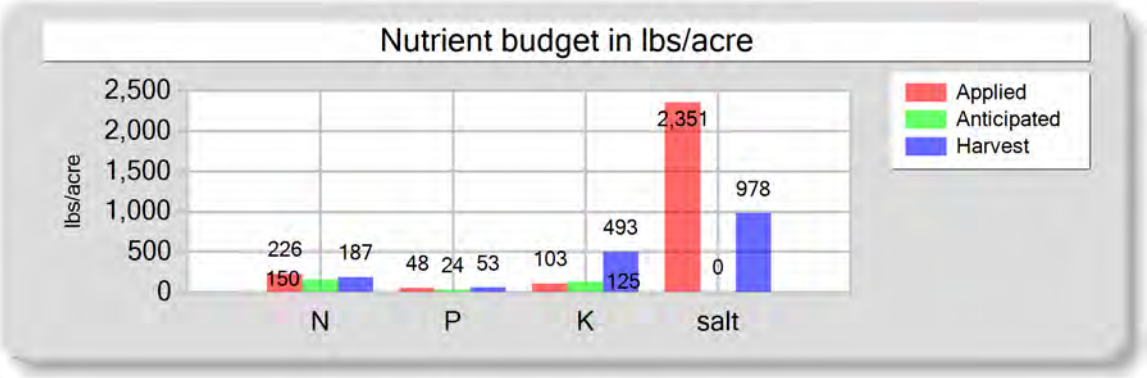
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.74	6.33	117.83	1,241.74	720,000.00 *gal*
TID Lateral 2.5	Surface water	0.00	0.00	0.00	26.81	5,508,000.00 *gal*
Application event totals		54.74	6.33	117.83	1,268.55	

B. NUTRIENT BUDGET

Christianson - 11/14/2022: Oats, silage-soft dough

Field name: Christianson Crop: Oats, silage-soft dough Plant date: 11/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	219.04	48.49	102.91	2,328.19
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	22.81
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	226.04	48.49	102.91	2,351.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	187.50	52.66	493.09	977.96
Nutrient balance	38.55	-4.16	-390.18	1,373.04
Applied to removed ratio	1.21	0.92	0.21	2.40

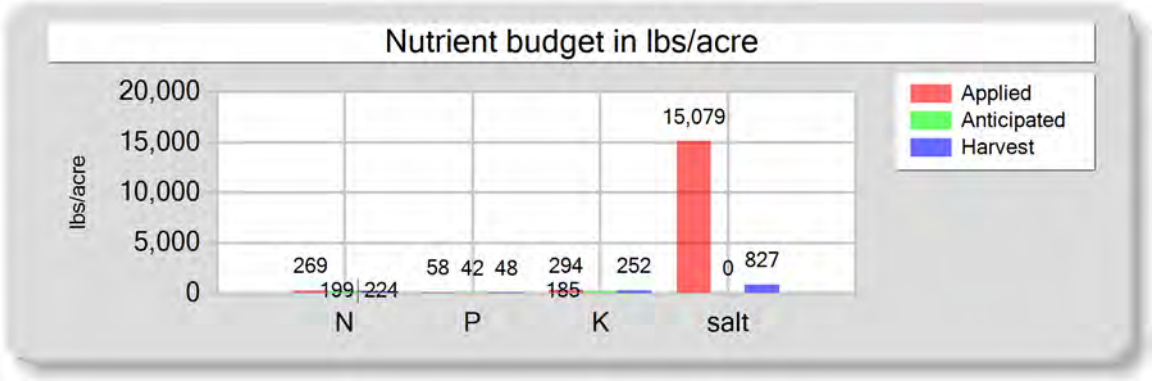
Fresh water applied	
10,125,000.00	*gallons*
372.87	*acre-inches*
11.65	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Christianson - 06/20/2023: Corn, silage

Field name: Christianson | Crop: Corn, silage | Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	16.00	0.00	0.00	0.00
Dry manure	245.60	57.70	293.68	14,891.91
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	187.26
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	268.60	57.70	293.68	15,079.17
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	199.24	47.91	252.05	826.91
Nutrient balance	69.36	9.79	41.63	14,252.26
Applied to removed ratio	1.35	1.20	1.17	18.24

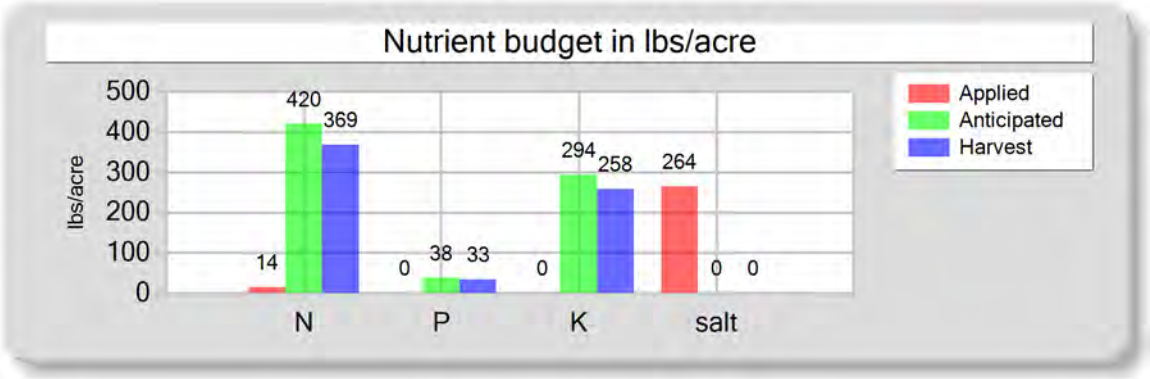
Fresh water applied	
39,892,500.00	*gallons*
1,469.11	*acre-inches*
45.91	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Coelho 1 - 11/18/2021: Alfalfa, hay

Field name: Coelho 1 Crop: Alfalfa, hay Plant date: 11/18/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	264.22
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	264.22
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	368.96	33.21	258.27	0.00
Nutrient balance	-354.96	-33.21	-258.27	264.22
Applied to removed ratio	0.04	0.00	0.00	0.00

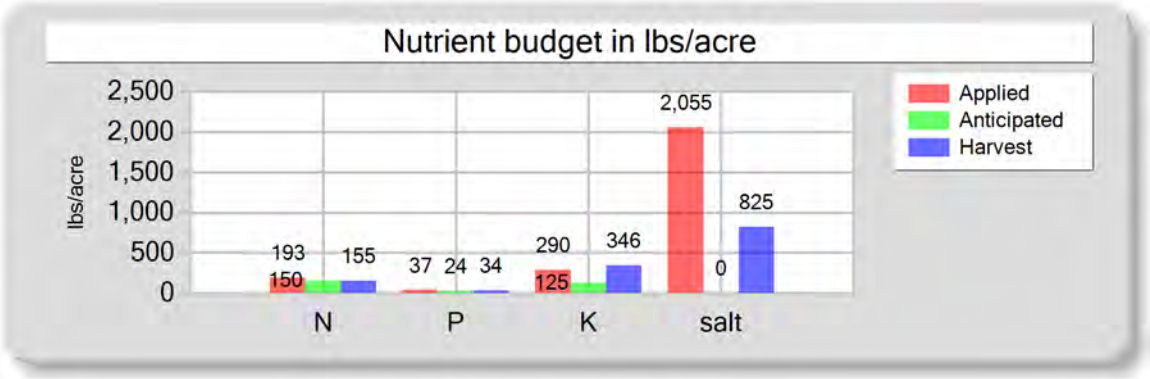
Fresh water applied	
78,570,000.00	*gallons*
2,893.47	*acre-inches*
43.19	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Coelho 2 - 11/11/2022: Oats, silage-soft dough

Field name: Coelho 2 Crop: Oats, silage-soft dough Plant date: 11/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	186.26	36.87	290.36	2,017.53
Fresh water	0.00	0.00	0.00	37.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	193.26	36.87	290.36	2,055.34
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	154.98	33.92	345.87	824.87
Nutrient balance	38.28	2.95	-55.52	1,230.47
Applied to removed ratio	1.25	1.09	0.84	2.49

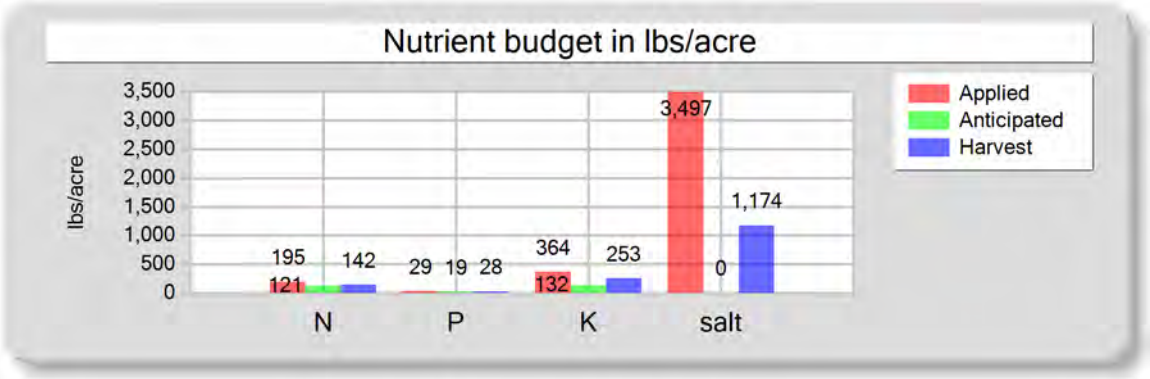
Fresh water applied	
11,947,500.00	*gallons*
439.99	*acre-inches*
12.57	*inches/acre*

Process wastewater applied	
1,770,000.00	*gallons*
65.18	*acre-inches*
1.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

Coelho 2 - 06/25/2023: Sudangrass, silage

Field name: Coelho 2 Crop: Sudangrass, silage Plant date: 06/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	188.09	29.48	364.14	3,356.12
Fresh water	0.00	0.00	0.00	140.79
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	195.09	29.48	364.14	3,496.91
Anticipated crop nutrient removal	121.00	18.70	132.00	0.00
Actual crop nutrient removal	141.58	28.36	252.96	1,173.57
Nutrient balance	53.52	1.12	111.18	2,323.34
Applied to removed ratio	1.38	1.04	1.44	2.98

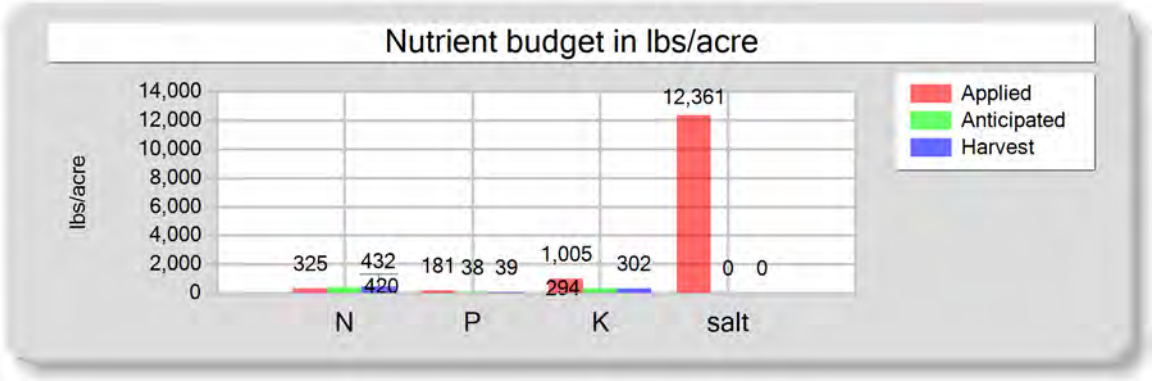
Fresh water applied	
21,870,000.00	*gallons*
805.40	*acre-inches*
23.01	*inches/acre*

Process wastewater applied	
2,400,000.00	*gallons*
88.38	*acre-inches*
2.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

Coelho 3 - 11/10/2022: Alfalfa, hay

Field name: Coelho 3 Crop: Alfalfa, hay Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	310.94	180.69	1,004.57	12,089.66
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	271.48
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	324.94	180.69	1,004.57	12,361.14
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	432.00	38.88	302.40	0.00
Nutrient balance	-107.06	141.81	702.17	12,361.14
Applied to removed ratio	0.75	4.65	3.32	0.00

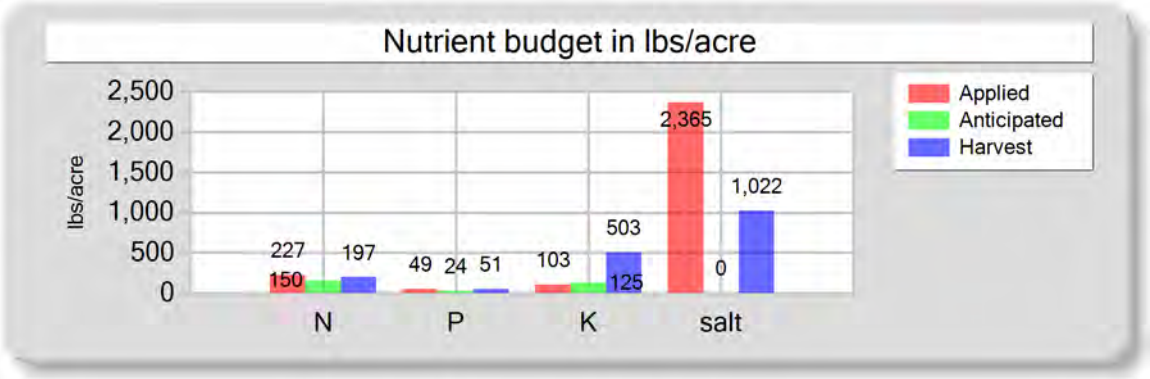
Fresh water applied	
54,675,000.00	*gallons*
2,013.49	*acre-inches*
50.34	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Coito-Keyes - 11/12/2022: Oats, silage-soft dough

Field name: Coito-Keyes Crop: Oats, silage-soft dough Plant date: 11/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	220.08	48.72	103.40	2,339.25
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	26.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	227.08	48.72	103.40	2,365.42
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	197.24	51.20	502.94	1,022.37
Nutrient balance	29.84	-2.48	-399.55	1,343.05
Applied to removed ratio	1.15	0.95	0.21	2.31

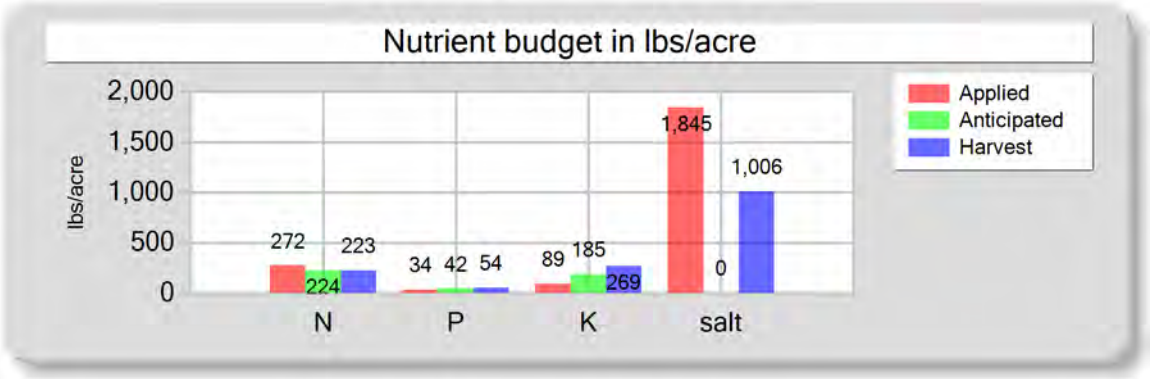
Fresh water applied	
14,580,000.00	*gallons*
536.93	*acre-inches*
11.42	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Coito-Keyes - 06/25/2023: Corn, silage

Field name: Coito-Keyes Crop: Corn, silage Plant date: 06/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	16.00	0.00	0.00	0.00
Dry manure	249.07	34.49	89.41	1,628.53
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	216.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	272.07	34.49	89.41	1,844.97
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	223.05	53.90	268.90	1,006.19
Nutrient balance	49.02	-19.42	-179.49	838.78
Applied to removed ratio	1.22	0.64	0.33	1.83

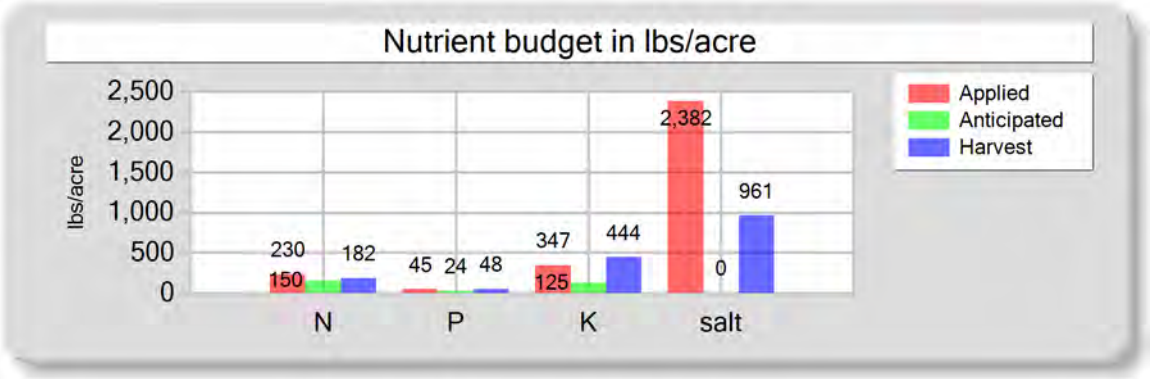
Fresh water applied	
58,050,000.00	*gallons*
2,137.78	*acre-inches*
45.48	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Coito-Vivan - 11/13/2022: Oats, silage-soft dough

Field name: Coito-Vivan Crop: Oats, silage-soft dough Plant date: 11/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	223.03	45.46	346.83	2,343.54
Fresh water	0.00	0.00	0.00	38.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	230.03	45.46	346.83	2,382.48
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	182.37	48.50	444.31	960.70
Nutrient balance	47.66	-3.04	-97.48	1,421.78
Applied to removed ratio	1.26	0.94	0.78	2.48

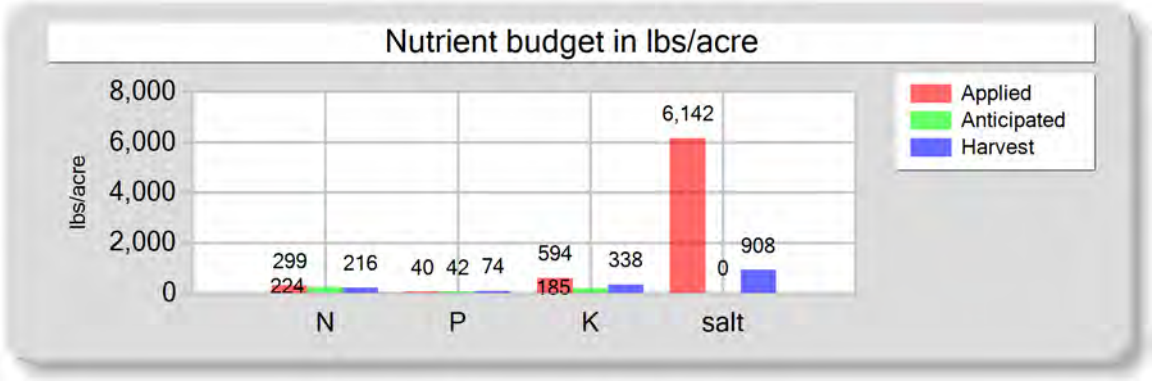
Fresh water applied	
25,920,000.00	*gallons*
954.55	*acre-inches*
12.73	*inches/acre*

Process wastewater applied	
4,800,000.00	*gallons*
176.77	*acre-inches*
2.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

Coito-Vivan - 06/17/2023: Corn, silage

Field name: Coito-Vivan Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	291.57	40.14	593.65	5,854.99
Fresh water	0.00	0.00	0.00	287.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	298.57	40.14	593.65	6,142.13
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	215.56	73.70	337.77	907.93
Nutrient balance	83.01	-33.55	255.88	5,234.19
Applied to removed ratio	1.39	0.54	1.76	6.76

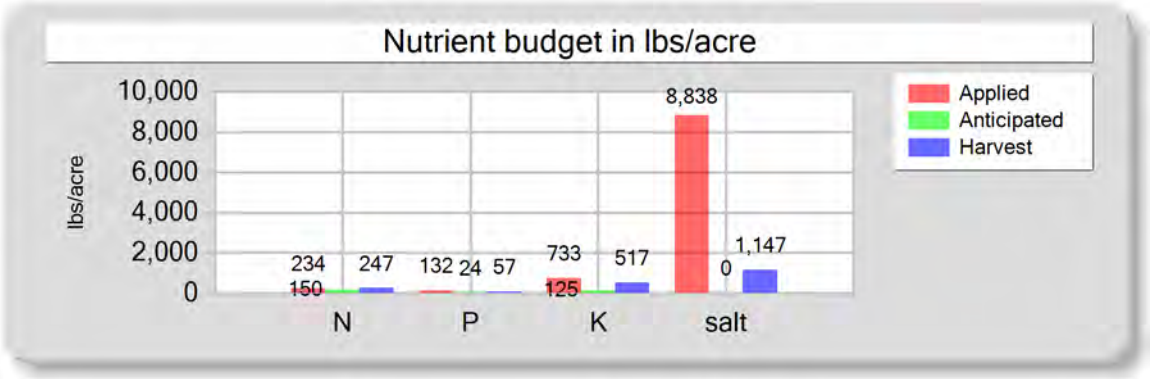
Fresh water applied	
95,580,000.00	*gallons*
3,519.89	*acre-inches*
46.93	*inches/acre*

Process wastewater applied	
7,980,000.00	*gallons*
293.88	*acre-inches*
3.92	*inches/acre*

Total harvests for the crop	
1	*harvests*

Gomes - 11/14/2022: Oats, silage-soft dough

Field name: Gomes | Crop: Oats, silage-soft dough | Plant date: 11/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	226.73	131.76	732.50	8,815.38
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	22.81
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	233.73	131.76	732.50	8,838.19
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	247.10	57.44	516.59	1,146.75
Nutrient balance	-13.37	74.32	215.91	7,691.44
Applied to removed ratio	0.95	2.29	1.42	7.71

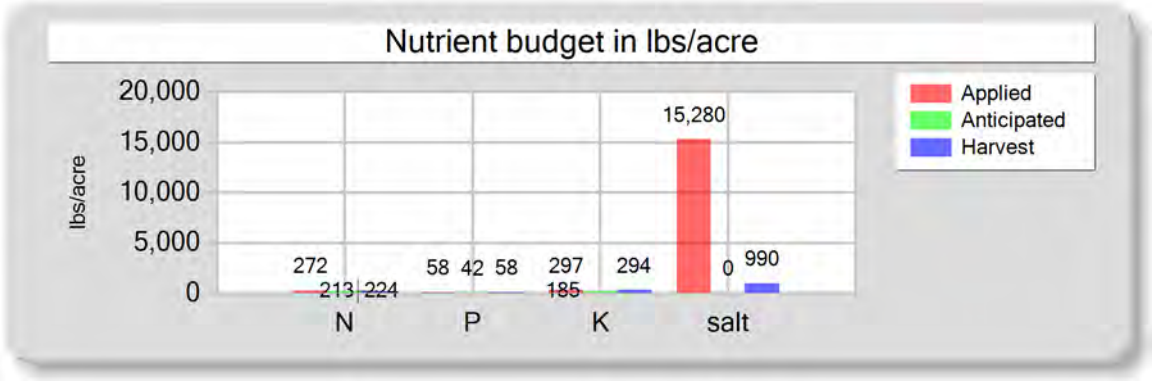
Fresh water applied
5,265,000.00 *gallons*
193.89 *acre-inches*
12.12 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Gomes - 06/20/2023: Corn, silage

Field name: Gomes Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	16.00	0.00	0.00	0.00
Dry manure	248.56	58.39	297.22	15,071.33
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	209.12
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	271.56	58.39	297.22	15,280.45
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	212.72	58.07	294.01	990.23
Nutrient balance	58.84	0.32	3.21	14,290.23
Applied to removed ratio	1.28	1.01	1.01	15.43

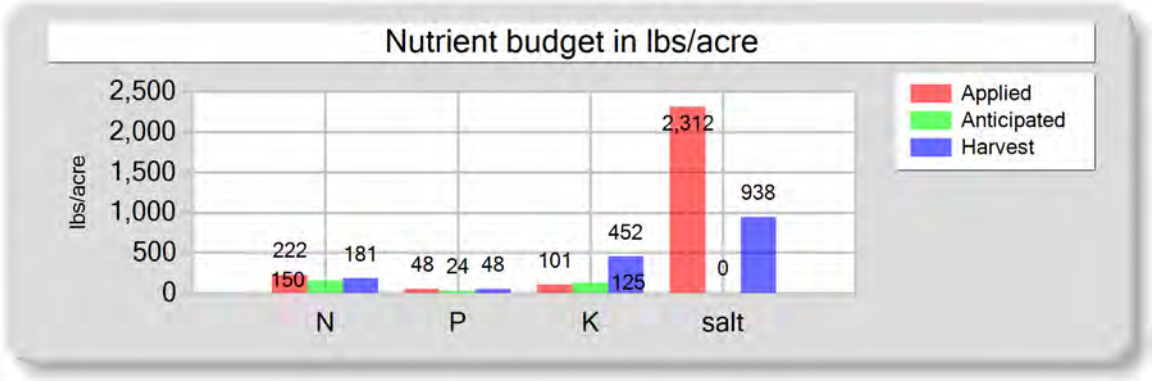
Fresh water applied	
22,275,000.00	*gallons*
820.31	*acre-inches*
51.27	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lacross - 11/14/2022: Oats, silage-soft dough

Field name: Lacross | Crop: Oats, silage-soft dough | Plant date: 11/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	214.90	47.58	100.96	2,284.17
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	28.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	221.90	47.58	100.96	2,312.18
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	181.04	47.66	451.83	938.45
Nutrient balance	40.86	-0.08	-350.87	1,373.74
Applied to removed ratio	1.23	1.00	0.22	2.46

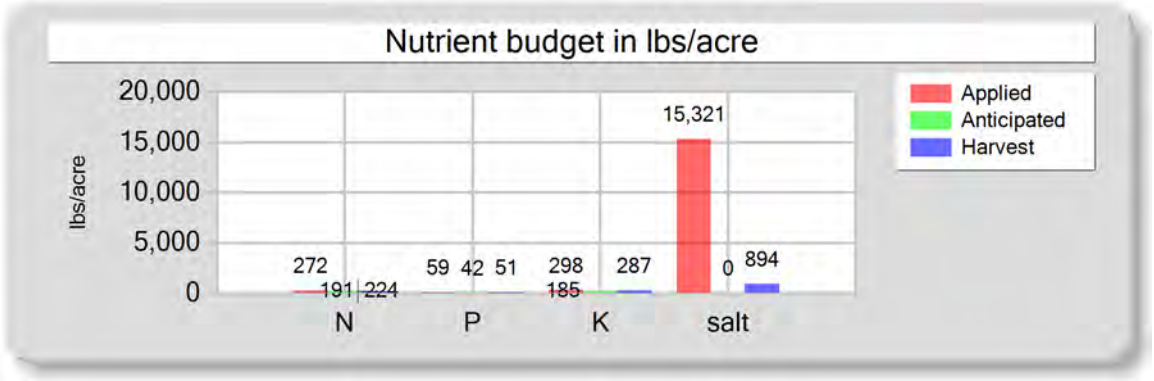
Fresh water applied	
12,555,000.00	*gallons*
462.36	*acre-inches*
12.17	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lacross - 06/21/2023: Corn, silage

Field name: Lacross Crop: Corn, silage Plant date: 06/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	16.00	0.00	0.00	0.00
Dry manure	249.18	58.54	297.96	15,109.11
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	211.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	272.18	58.54	297.96	15,321.09
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	190.99	51.09	287.36	893.55
Nutrient balance	81.19	7.45	10.60	14,427.55
Applied to removed ratio	1.43	1.15	1.04	17.15

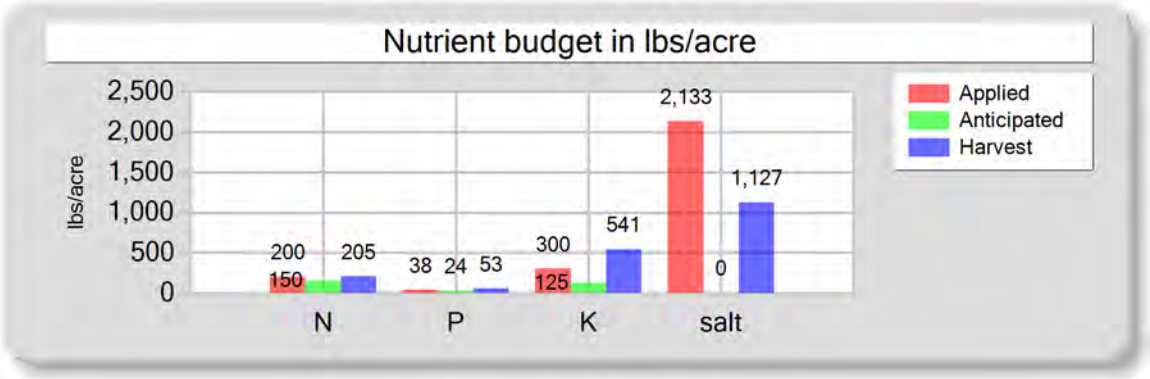
Fresh water applied	
45,967,500.00	*gallons*
1,692.83	*acre-inches*
44.55	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

M1 - 11/15/2022: Oats, silage-soft dough

Field name: M1 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	192.54	37.94	300.26	2,095.20
Fresh water	0.00	0.00	0.00	37.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	199.54	37.94	300.26	2,132.53
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	204.92	53.20	541.41	1,127.33
Nutrient balance	-5.38	-15.25	-241.15	1,005.21
Applied to removed ratio	0.97	0.71	0.55	1.89

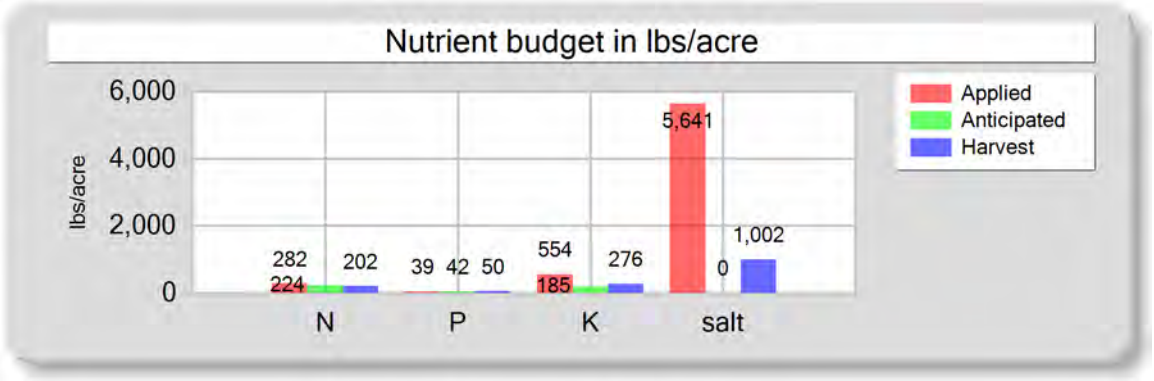
Fresh water applied	
30,780,000.00	*gallons*
1,133.52	*acre-inches*
12.88	*inches/acre*

Process wastewater applied	
4,560,000.00	*gallons*
167.93	*acre-inches*
1.91	*inches/acre*

Total harvests for the crop	
1	*harvests*

M1 - 06/27/2023: Corn, silage

Field name: M1 Crop: Corn, silage Plant date: 06/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	274.70	38.78	554.26	5,403.46
Fresh water	0.00	0.00	0.00	237.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	281.70	38.78	554.26	5,640.93
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	202.21	49.58	275.74	1,001.57
Nutrient balance	79.49	-10.79	278.53	4,639.36
Applied to removed ratio	1.39	0.78	2.01	5.63

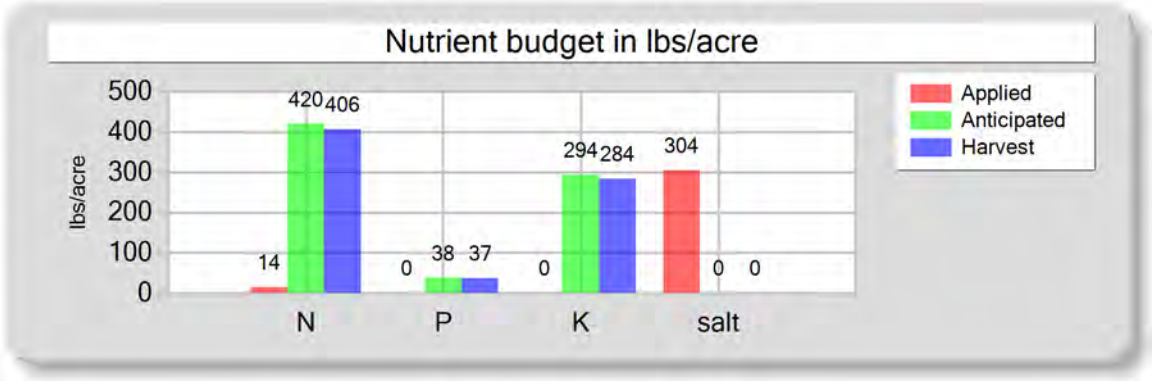
Fresh water applied	
92,745,000.00	*gallons*
3,415.48	*acre-inches*
38.81	*inches/acre*

Process wastewater applied	
8,820,000.00	*gallons*
324.81	*acre-inches*
3.69	*inches/acre*

Total harvests for the crop	
1	*harvests*

M2 - 11/22/2018: Alfalfa, hay

Field name: M2 Crop: Alfalfa, hay Plant date: 11/22/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	304.18
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	304.18
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	406.32	36.57	284.42	0.00
Nutrient balance	-392.32	-36.57	-284.42	304.18
Applied to removed ratio	0.03	0.00	0.00	0.00

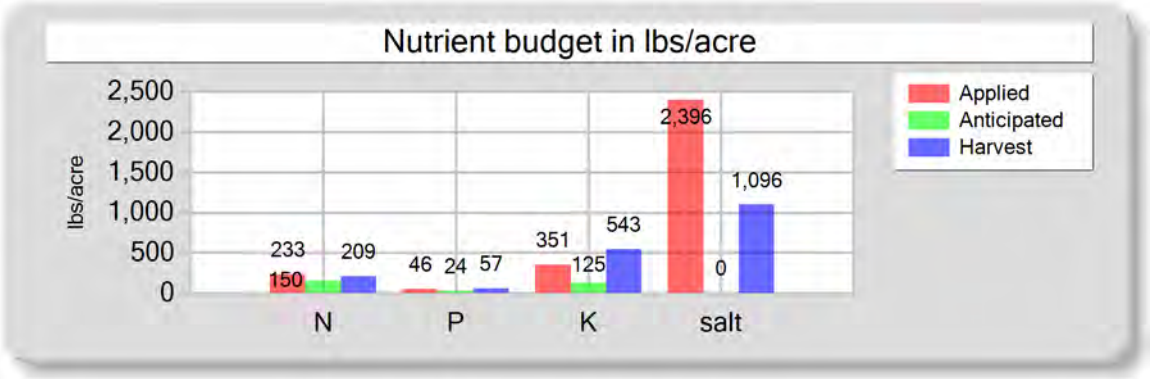
Fresh water applied	
76,950,000.00	*gallons*
2,833.81	*acre-inches*
49.72	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Machado - 11/16/2022: Oats, silage-soft dough

Field name: Machado | Crop: Oats, silage-soft dough | Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	225.89	46.21	351.18	2,364.36
Fresh water	0.00	0.00	0.00	32.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	232.89	46.21	351.18	2,396.38
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	209.28	57.04	543.45	1,096.07
Nutrient balance	23.62	-10.83	-192.27	1,300.30
Applied to removed ratio	1.11	0.81	0.65	2.19

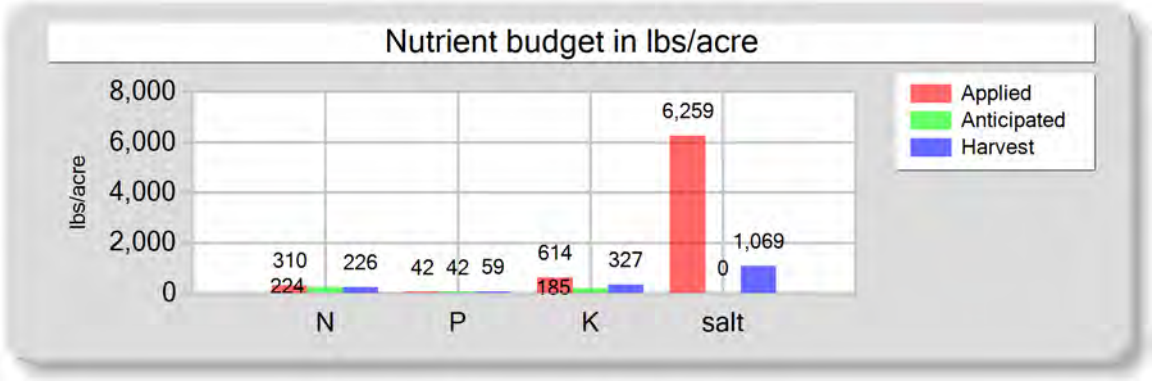
Fresh water applied	
17,820,000.00	*gallons*
656.25	*acre-inches*
11.51	*inches/acre*

Process wastewater applied	
3,720,000.00	*gallons*
136.99	*acre-inches*
2.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

Machado - 06/16/2023: Corn, silage

Field name: Machado | Crop: Corn, silage | Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	302.90	42.13	614.46	6,032.12
Fresh water	0.00	0.00	0.00	226.53
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	309.90	42.13	614.46	6,258.65
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	226.36	59.43	327.18	1,069.42
Nutrient balance	83.54	-17.29	287.28	5,189.23
Applied to removed ratio	1.37	0.71	1.88	5.85

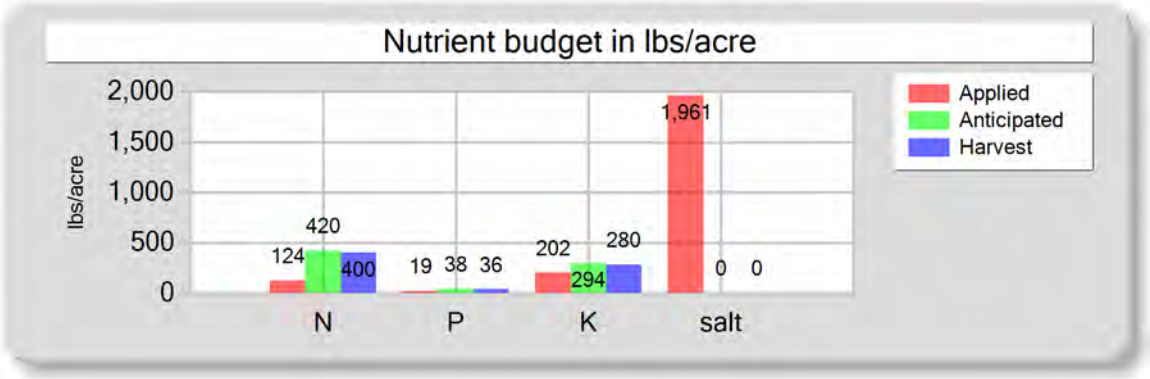
Fresh water applied	
57,307,500.00	*gallons*
2,110.44	*acre-inches*
37.03	*inches/acre*

Process wastewater applied	
6,300,000.00	*gallons*
232.01	*acre-inches*
4.07	*inches/acre*

Total harvests for the crop	
1	*harvests*

Morris 1 - 11/28/2020: Alfalfa, hay

Field name: Morris 1 Crop: Alfalfa, hay Plant date: 11/28/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	109.82	19.14	202.45	1,732.42
Fresh water	0.00	0.00	0.00	228.69
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	123.82	19.14	202.45	1,961.12
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	400.00	36.00	280.00	0.00
Nutrient balance	-276.18	-16.86	-77.55	1,961.12
Applied to removed ratio	0.31	0.53	0.72	0.00

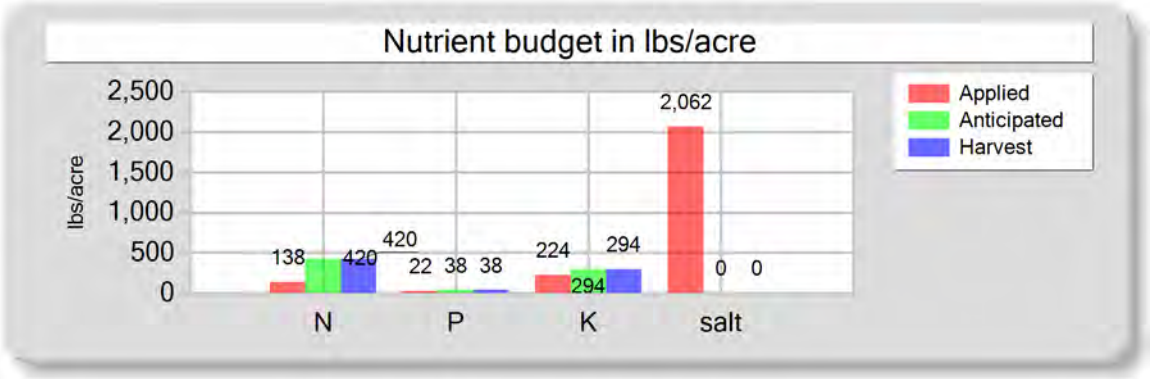
Fresh water applied	
3,915,000.00	*gallons*
144.18	*acre-inches*
48.06	*inches/acre*

Process wastewater applied	
120,000.00	*gallons*
4.42	*acre-inches*
1.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

Morris 2 - 11/28/2022: Alfalfa, hay

Field name: Morris 2 Crop: Alfalfa, hay Plant date: 11/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	123.59	22.35	223.60	1,855.09
Fresh water	0.00	0.00	0.00	207.01
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	137.59	22.35	223.60	2,062.10
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	420.00	37.80	294.00	0.00
Nutrient balance	-282.41	-15.45	-70.40	2,062.10
Applied to removed ratio	0.33	0.59	0.76	0.00

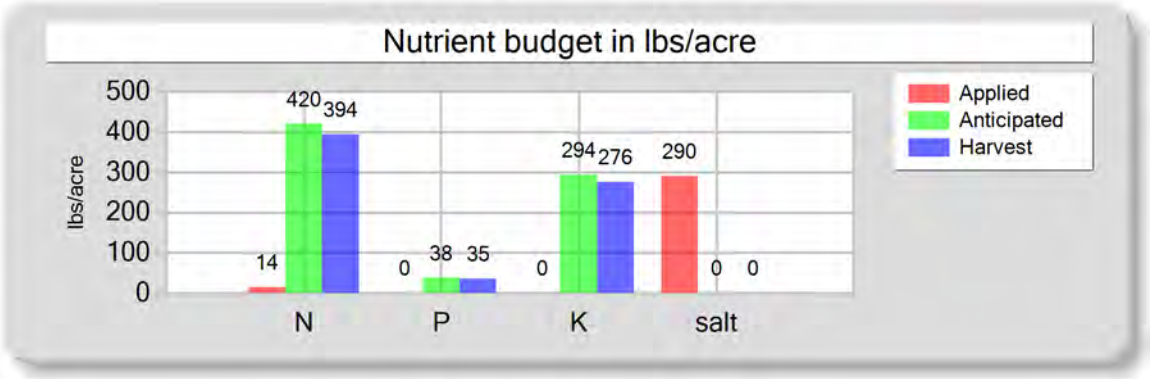
Fresh water applied	
14,175,000.00	*gallons*
522.02	*acre-inches*
43.50	*inches/acre*

Process wastewater applied	
540,000.00	*gallons*
19.89	*acre-inches*
1.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

Soares - 11/23/2018: Alfalfa, hay

Field name: Soares Crop: Alfalfa, hay Plant date: 11/23/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	290.35
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	290.35
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	394.29	35.49	276.00	0.00
Nutrient balance	-380.29	-35.49	-276.00	290.35
Applied to removed ratio	0.04	0.00	0.00	0.00

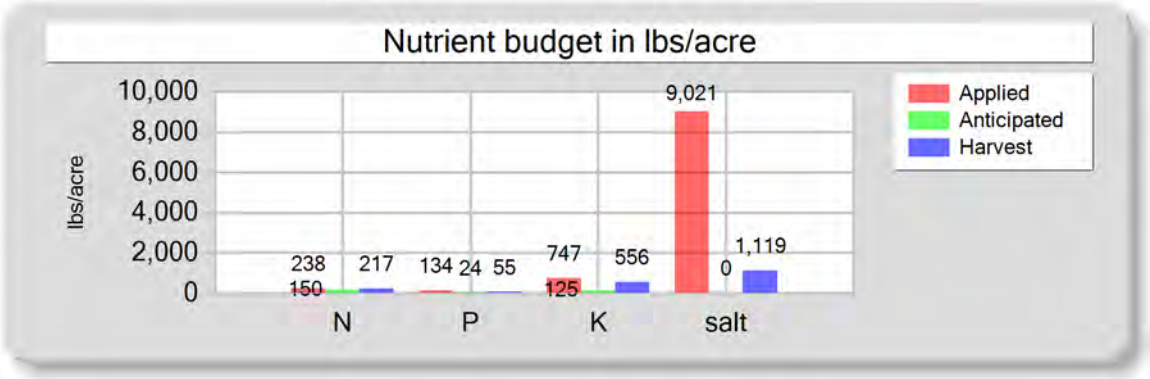
Fresh water applied	
99,225,000.00	*gallons*
3,654.12	*acre-inches*
47.46	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Sousa - 11/14/2022: Oats, silage-soft dough

Field name: Sousa | Crop: Oats, silage-soft dough | Plant date: 11/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	231.35	134.44	747.45	8,995.29
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	26.07
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	238.35	134.44	747.45	9,021.36
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	217.42	54.60	555.89	1,119.48
Nutrient balance	20.93	79.84	191.56	7,901.88
Applied to removed ratio	1.10	2.46	1.34	8.06

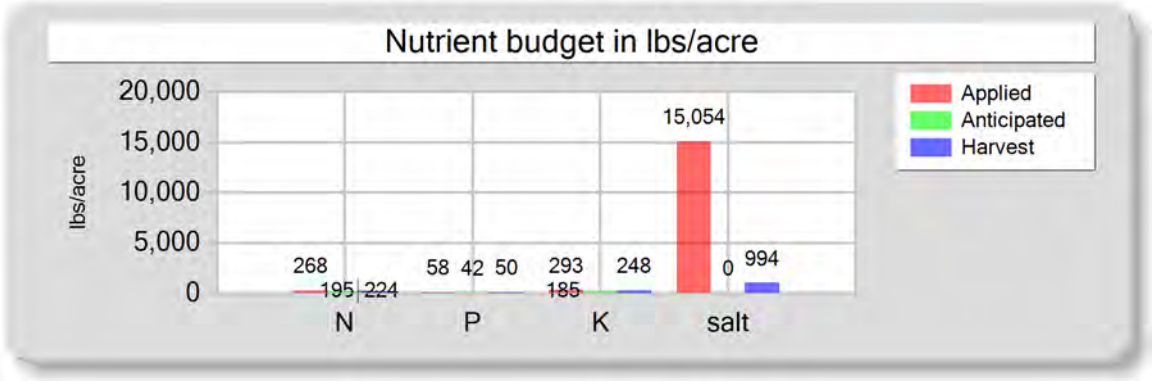
Fresh water applied	
19,440,000.00	*gallons*
715.91	*acre-inches*
12.78	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Sousa - 06/20/2023: Corn, silage

Field name: Sousa | Crop: Corn, silage | Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	16.00	0.00	0.00	0.00
Dry manure	245.18	57.60	293.17	14,866.28
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	187.39
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	268.18	57.60	293.17	15,053.67
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	195.12	50.45	248.01	993.97
Nutrient balance	73.05	7.14	45.17	14,059.71
Applied to removed ratio	1.37	1.14	1.18	15.15

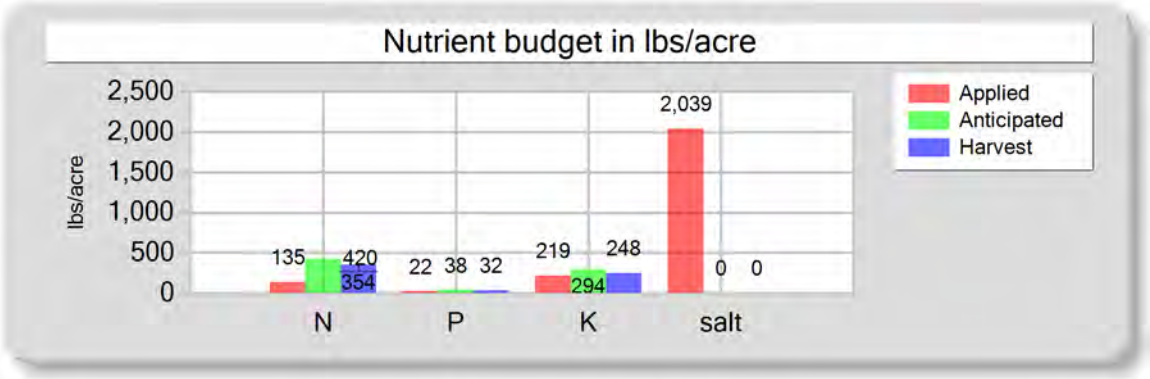
Fresh water applied	
69,862,500.00	*gallons*
2,572.80	*acre-inches*
45.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Sousa Carpenter - 12/08/2019: Alfalfa, hay

Field name: Sousa Carpenter Crop: Alfalfa, hay Plant date: 12/08/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	120.84	21.71	219.37	1,830.56
Fresh water	0.00	0.00	0.00	208.19
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	134.84	21.71	219.37	2,038.75
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	354.00	31.86	247.80	0.00
Nutrient balance	-219.16	-10.15	-28.43	2,038.75
Applied to removed ratio	0.38	0.68	0.89	0.00

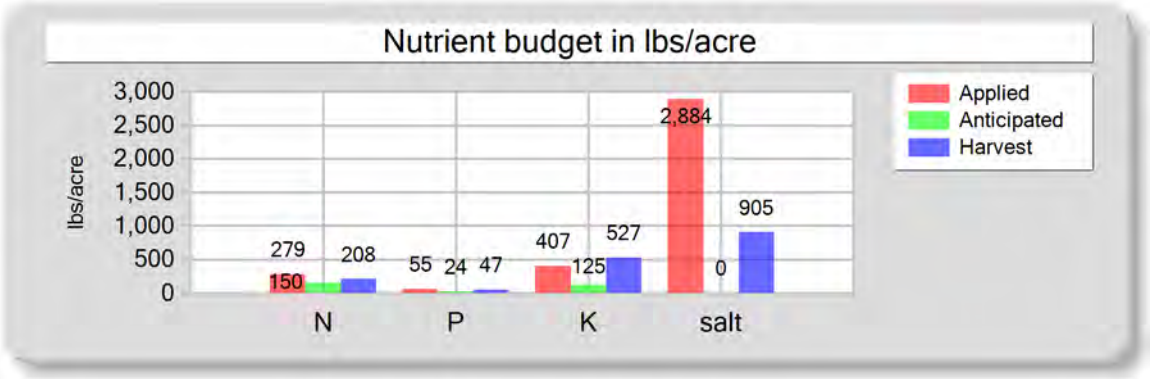
Fresh water applied	
106,920,000.00	*gallons*
3,937.50	*acre-inches*
43.75	*inches/acre*

Process wastewater applied	
3,960,000.00	*gallons*
145.83	*acre-inches*
1.62	*inches/acre*

Total harvests for the crop	
1	*harvests*

Sung-Ken - 11/17/2022: Oats, silage-soft dough

Field name: Sung-Ken Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	262.34	55.26	406.80	2,657.55
Fresh water	9.65	0.00	0.00	226.72
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	278.99	55.26	406.80	2,884.27
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	207.55	47.38	527.34	904.99
Nutrient balance	71.44	7.89	-120.54	1,979.28
Applied to removed ratio	1.34	1.17	0.77	3.19

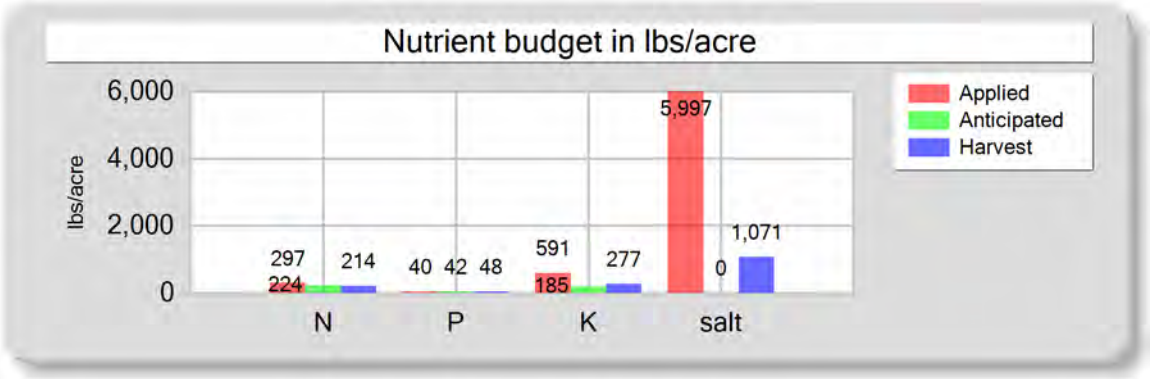
Fresh water applied	
20,088,000.00	*gallons*
739.77	*acre-inches*
12.33	*inches/acre*

Process wastewater applied	
4,800,000.00	*gallons*
176.77	*acre-inches*
2.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

Sung-Ken - 06/15/2023: Corn, silage

Field name: Sung-Ken Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	290.47	40.02	591.29	5,830.15
Fresh water	0.00	0.00	0.00	166.79
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.47	40.02	591.29	5,996.94
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	213.99	47.55	276.79	1,071.06
Nutrient balance	83.48	-7.54	314.50	4,925.88
Applied to removed ratio	1.39	0.84	2.14	5.60

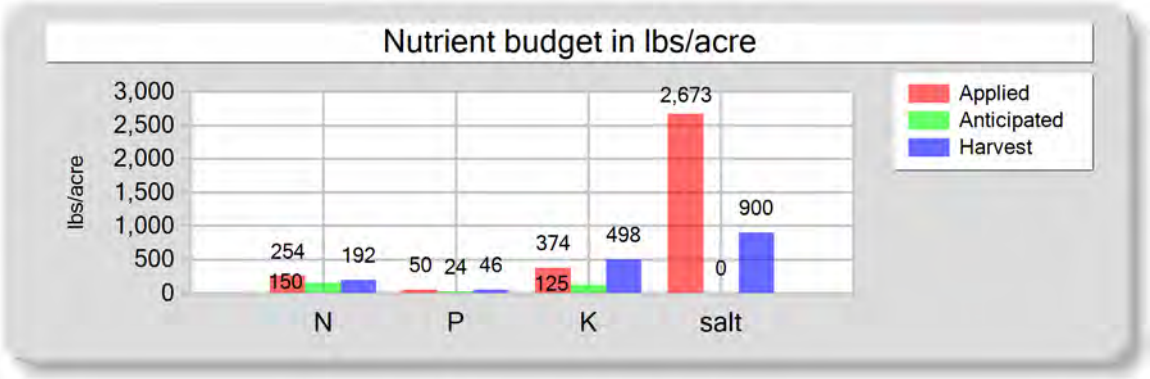
Fresh water applied	
57,105,000.00	*gallons*
2,102.98	*acre-inches*
35.05	*inches/acre*

Process wastewater applied	
6,360,000.00	*gallons*
234.22	*acre-inches*
3.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

Van Rys 1 - 11/17/2022: Oats, silage-soft dough

Field name: Van Rys 1 Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	241.10	50.36	374.14	2,466.31
Fresh water	8.16	0.00	0.00	206.43
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	253.92	50.36	374.14	2,672.74
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	192.40	45.87	498.35	900.01
Nutrient balance	61.52	4.49	-124.21	1,772.73
Applied to removed ratio	1.32	1.10	0.75	2.97

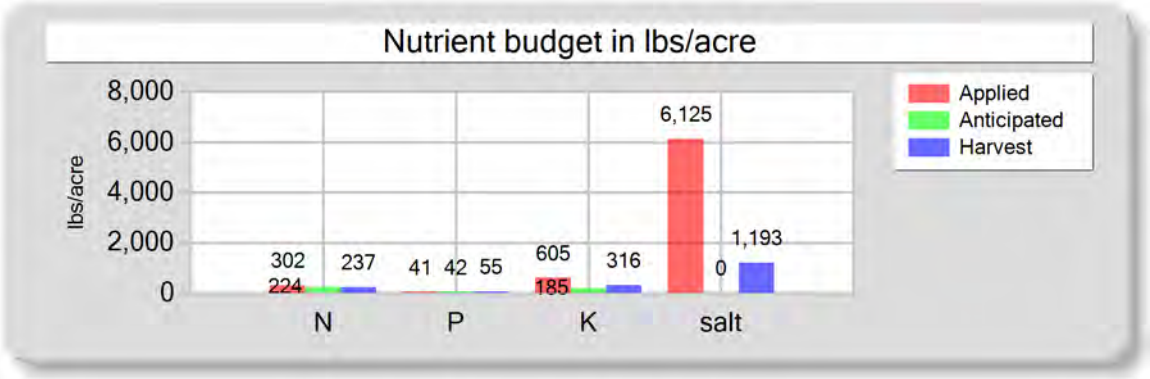
Fresh water applied	
23,094,000.00	*gallons*
850.47	*acre-inches*
13.50	*inches/acre*

Process wastewater applied	
4,560,000.00	*gallons*
167.93	*acre-inches*
2.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

Van Rys 1 - 05/16/2023: Corn, silage

Field name: Van Rys 1 Crop: Corn, silage Plant date: 05/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	297.53	41.14	604.86	5,954.04
Fresh water	0.00	0.00	0.00	171.09
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	302.19	41.14	604.86	6,125.13
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	236.90	55.32	316.27	1,192.92
Nutrient balance	65.29	-14.18	288.59	4,932.21
Applied to removed ratio	1.28	0.74	1.91	5.13

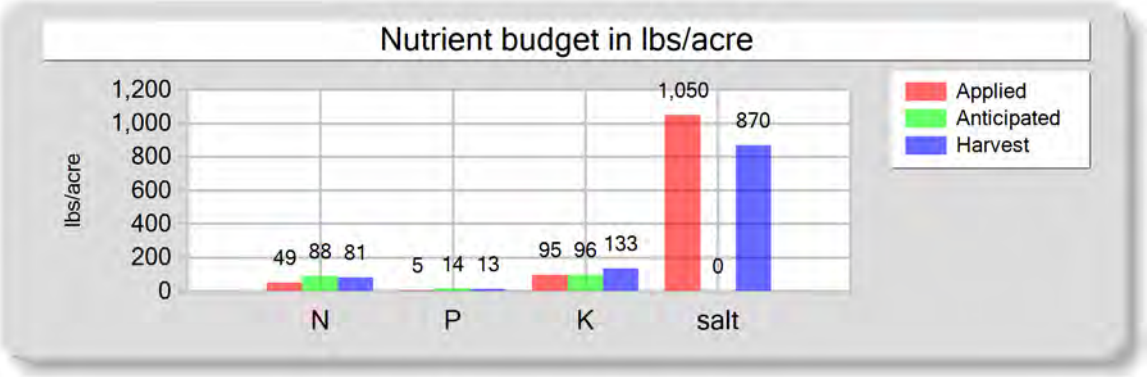
Fresh water applied	
61,506,000.00	*gallons*
2,265.06	*acre-inches*
35.95	*inches/acre*

Process wastewater applied	
6,840,000.00	*gallons*
251.89	*acre-inches*
4.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Van Rys 1 - 09/02/2023: Sudangrass, silage

Field name: Van Rys 1 Crop: Sudangrass, silage Plant date: 09/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	44.32	5.12	95.39	1,005.21
Fresh water	0.00	0.00	0.00	44.69
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	48.98	5.12	95.39	1,049.90
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	81.09	13.21	133.41	869.83
Nutrient balance	-32.11	-8.09	-38.02	180.07
Applied to removed ratio	0.60	0.39	0.71	1.21

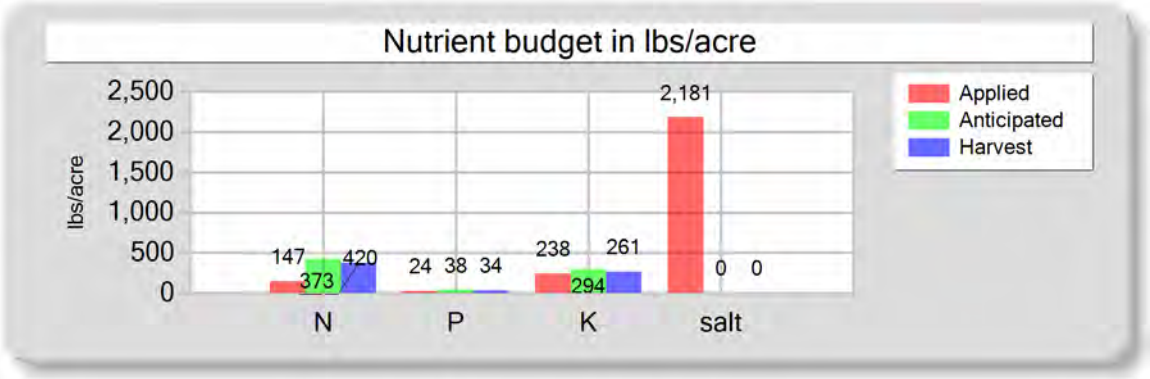
Fresh water applied	
16,065,000.00	*gallons*
591.62	*acre-inches*
9.39	*inches/acre*

Process wastewater applied	
1,020,000.00	*gallons*
37.56	*acre-inches*
0.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

Van Rys 2 - 11/28/2020: Alfalfa, hay

Field name: Van Rys 2 Crop: Alfalfa, hay Plant date: 11/28/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	132.77	24.48	237.71	1,936.87
Fresh water	0.00	0.00	0.00	244.07
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	146.77	24.48	237.71	2,180.95
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	373.33	33.60	261.33	0.00
Nutrient balance	-226.56	-9.12	-23.63	2,180.95
Applied to removed ratio	0.39	0.73	0.91	0.00

Fresh water applied	
50,139,000.00	*gallons*
1,846.45	*acre-inches*
51.29	*inches/acre*

Process wastewater applied	
1,740,000.00	*gallons*
64.08	*acre-inches*
1.78	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 11/15/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 39.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	8,310.00	46,200.00							55.60
DL	100.00	100.00	100.00							0.01

Separated Solids

Sample and source description: Separated Solids

Sample date: 11/15/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	3,520.00	7,470.00	12,900.00	4,970.00	2,200.00	3,400.00	860.00		16.90
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 05/03/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 49.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,450.00	2,220.00	11,300.00							57.30
DL	100.00	100.00	100.00							0.01

Sep. Solids

Sample and source description: Sep. Solids

Sample date: 05/03/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,600.00	2,160.00	5,600.00							10.20
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/23/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 32.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,700.00	4,510.00	15,700.00							55.90
DL	100.00	100.00	100.00							0.01

Separated Solids

Sample and source description: Separated Solids

Sample date: 10/23/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,600.00	2,590.00	3,630.00	7,640.00	3,360.00	1,040.00	2,370.00	496.00		6.30
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWSQ4

Sample and source description: WWSQ4

Sample date: 12/13/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	549.00	423.00	0.00	0.00	97.60	863.00								7,520.00	6,560
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	330.00	193.00	0.00	0.00	76.80	507.00								4,610.00	2,940
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	14.00	2.80	0.00	0.00	3.48	33.80								1,840.00	2,460
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	328.00	227.00	0.00	0.00	37.90	706.00	135.00	106.00	378.00	2,500.00	0.00	168.00	137.00	6,020.00	7,440
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

C. FRESH WATER ANALYSES

IW #1 Dairy

IW #1 Dairy

Sample description: IW #1 Dairy

Sample date: 08/29/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	31.10	0.00	31.10								1,080.00	680
DL	1.00	1.00	0.10								10.00	10

IW #2 Carpenter

IW #2 Carpenter

Sample description: IW #2 Carpenter

Sample date: 08/29/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	30.60	0.00	30.00								997.00	640
DL	1.00	1.00	0.10								10.00	10

IW #3 Vivian

IW #3 Vivian

Sample description: IW #3 Vivian

Sample date: 08/29/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	55.40	0.00	55.00								1,710.00	1,100
DL	1.00	1.00	0.10								10.00	10

TID Lateral 2

TID Lateral 2

Sample description: TID Lateral 2

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	0.50	0.50	0.50								1.00	10

TID Lateral 2

Sample description: TID Lateral 2

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.30	18
DL	1.00	1.00	0.10								10.00	10

TID Lateral 2.5

TID Lateral 2.5

Sample description: TID Lateral 2.5

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	0.50	0.50	0.50								1.00	10

TID Lateral 2.5

TID Lateral 2.5

Sample description: TID Lateral 2.5

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.50	21
DL	1.00	1.00	0.10								10.00	10

TID Lateral 3

TID Lateral 3

Sample description: TID Lateral 3

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	0.50	0.50	0.50								1.00	10

TID Lateral 3

Sample description: TID Lateral 3

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								35.90	27
DL	1.00	1.00	0.10								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Christianson - 11/14/2022: Oats, silage-soft dough

Christianson

Sample and source description: Christianson

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,160.00	1,730.00	16,200.00		11.90
DL	100.00	100.00	100.00		0.01

Christianson - 06/20/2023: Corn, silage

Christianson

Sample and source description: Christianson

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,660.00	880.00	4,630.00		4.90
DL	100.00	100.00	100.00		0.01

Coelho 2 - 11/11/2022: Oats, silage-soft dough

Coelho 2

Sample and source description: Coelho 2

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,990.00	1,530.00	15,600.00		13.10
DL	100.00	100.00	100.00		0.01

Coelho 2 - 06/25/2023: Sudangrass, silage

Coelho 2

Sample and source description: Coelho 2

Sample date: 11/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,940.00	1,390.00	12,400.00		15.30
DL	100.00	100.00	100.00		0.01

Coito-Keyes - 11/12/2022: Oats, silage-soft dough

Coito-Keyes

Sample and source description: Coito-Keyes

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,510.00	1,690.00	16,600.00		11.40
DL	100.00	100.00	100.00		0.01

Coito-Keyes - 06/25/2023: Corn, silage

Coito Keyes

Sample and source description: Coito Keyes

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,600.00	870.00	4,340.00		5.60
DL	100.00	100.00	100.00		0.01

Coito-Vivan - 11/13/2022: Oats, silage-soft dough

Coito-Vivian

Sample and source description: Coito-Vivian

Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,280.00	1,670.00	15,300.00		11.90
DL	100.00	100.00	100.00		0.01

Coito-Vivan - 06/17/2023: Corn, silage

Coito Vivian

Sample and source description: Coito Vivian

Sample date: 10/14/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,510.00	1,200.00	5,500.00		4.80
DL	100.00	100.00	100.00		0.01

Gomes - 11/14/2022: Oats, silage-soft dough

Gomes

Sample and source description: Gomes

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,840.00	1,590.00	14,300.00		12.40
DL	100.00	100.00	100.00		0.01

Gomes - 06/20/2023: Corn, silage

Gomes

Sample and source description: Gomes

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,480.00	950.00	4,810.00		5.40
DL	100.00	100.00	100.00		0.01

Lacross - 11/14/2022: Oats, silage-soft dough

La Cross

Sample and source description: La Cross

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,850.00	1,540.00	14,600.00		11.40
DL	100.00	100.00	100.00		0.01

Lacross - 06/21/2023: Corn, silage

La Cross

Sample and source description: La Cross

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,290.00	880.00	4,950.00		5.20
DL	100.00	100.00	100.00		0.01

M1 - 11/15/2022: Oats, silage-soft dough

Marshall

Sample and source description: Marshall

Sample date: 05/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,510.00	1,690.00	17,200.00		12.70
DL	100.00	100.00	100.00		0.01

M1 - 06/27/2023: Corn, silage

Marshall

Sample and source description: Marshall

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,630.00	890.00	4,950.00		5.80
DL	100.00	100.00	100.00		0.01

Machado - 11/16/2022: Oats, silage-soft dough

Machado

Sample and source description: Machado

Sample date: 04/29/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,200.00	1,690.00	16,100.00		12.30
DL	100.00	100.00	100.00		0.01

Machado - 06/16/2023: Corn, silage

Machado

Sample and source description: Machado

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,390.00	890.00	4,900.00		5.20
DL	100.00	100.00	100.00		0.01

Sousa - 11/14/2022: Oats, silage-soft dough

Sousa

Sample and source description: Sousa

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,610.00	1,660.00	16,900.00		11.90
DL	100.00	100.00	100.00		0.01

Sousa - 06/20/2023: Corn, silage

Sousa

Sample and source description: Sousa

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,210.00	830.00	4,080.00		5.60
DL	100.00	100.00	100.00		0.01

Sung-Ken - 11/17/2022: Oats, silage-soft dough

Sung-Ken

Sample and source description: Sung-Ken

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,360.00	1,680.00	18,700.00		11.30
DL	100.00	100.00	100.00		0.01

Sung-Ken - 06/15/2023: Corn, silage

Sung Ken

Sample and source description: Sung Ken

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,510.00	780.00	4,540.00		6.10
DL	100.00	100.00	100.00		0.01

Van Rys 1 - 11/17/2022: Oats, silage-soft dough

Van Rys 1

Sample and source description: Van Rys 1

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,760.00	1,850.00	20,100.00		11.00
DL	100.00	100.00	100.00		0.01

Van Rys 1 - 05/16/2023: Corn, silage

Van Rys 1

Sample and source description: Van Rys 1

Sample date: 08/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,940.00	920.00	5,260.00		6.20
DL	100.00	100.00	100.00		0.01

Van Rys 1 - 09/02/2023: Sudangrass, silage

Van Rys 1

Sample and source description: Van Rys 1

Sample date: 11/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,200.00	1,010.00	10,200.00		20.40
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

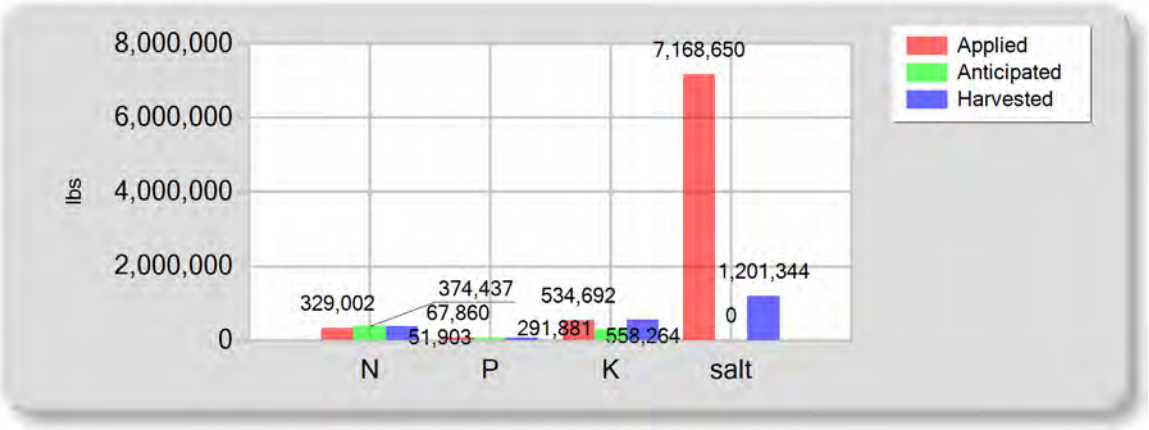
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

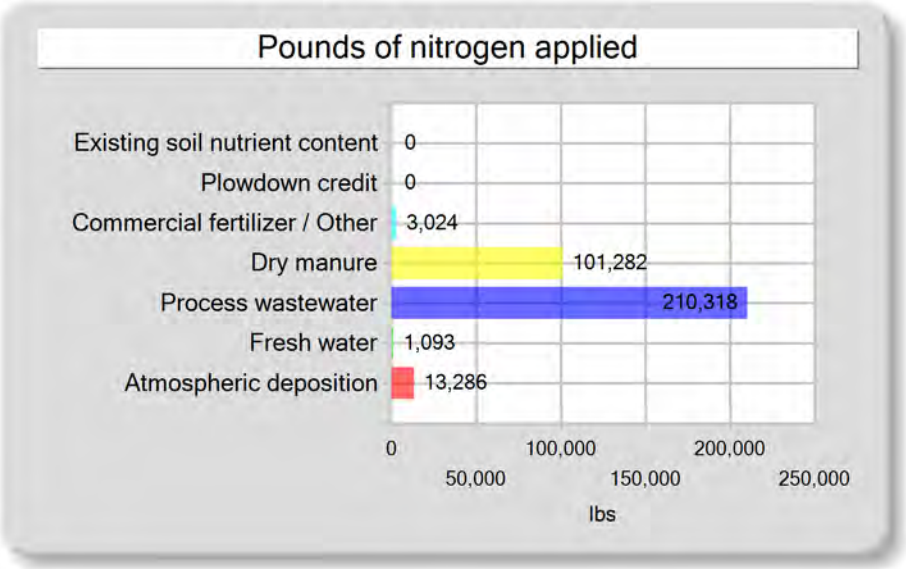
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

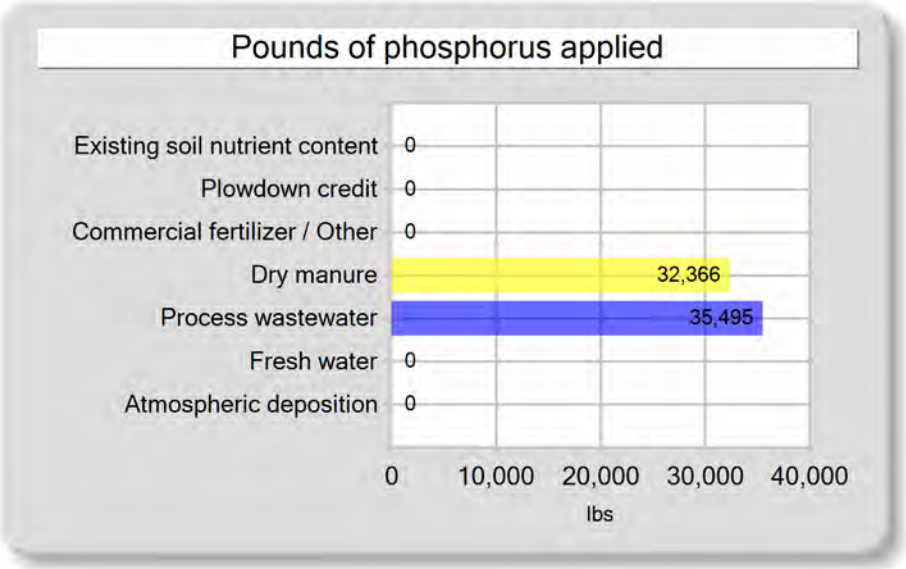
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	3,024.00	0.00	0.00	0.00
Dry manure	101,281.66	32,365.54	151,845.34	3,600,494.64
Process wastewater	210,317.53	35,494.67	382,846.99	3,306,457.57
Fresh water	1,093.02	0.00	0.00	261,697.33
Atmospheric deposition	13,286.00	0.00	0.00	0.00
Total nutrients applied	329,002.21	67,860.21	534,692.32	7,168,649.54
Anticipated crop nutrient removal	374,437.00	51,902.90	291,881.10	0.00
Actual crop nutrient removal	382,115.50	72,806.74	558,263.64	1,201,344.16
Nutrient balance	-53,113.29	-4,946.53	-23,571.32	5,967,305.37
Applied to removed ratio	0.86	0.93	0.96	5.97

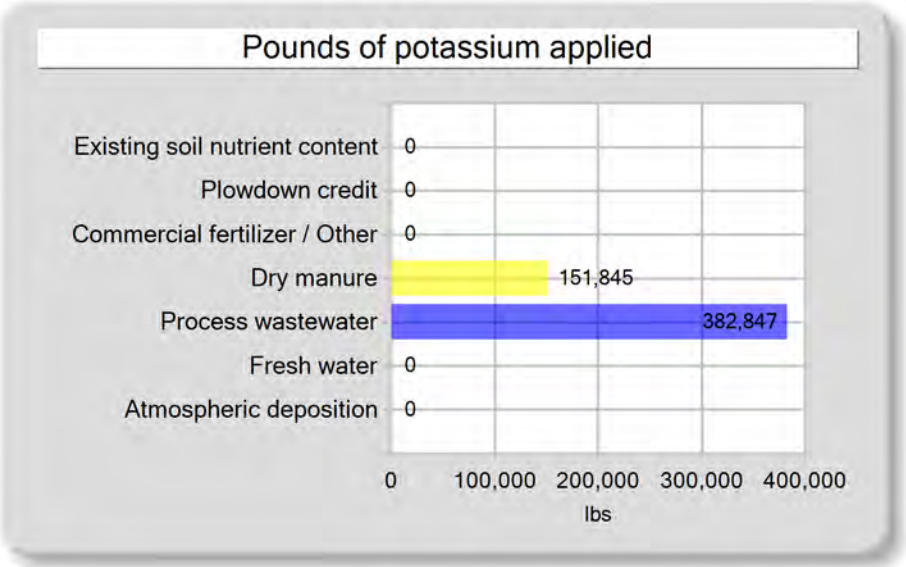
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

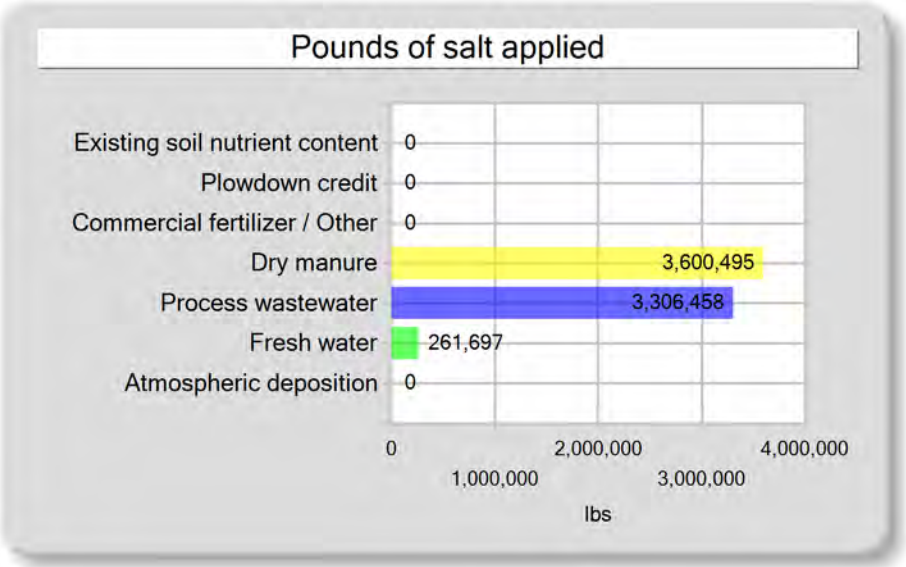


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

nancy Azevedo (Jun 25, 2024 10:48 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Tony J. & Delfina R Morris	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 25, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Tony Morris

Name of Dairy Facility: Tony Morris Dairy

Facility Address:

Number and Street	City	County	Zip Code
6506 Vivian RD	Modesto	Stanislaus	95358

Contact Person Name and Phone Number:

Name	Phone Number
Tony Morris	(209) 678-2300

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Tony Morris Dairy

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
P.O. Box 580596	Modesto	CA	95358

Contact Person:

Name	Phone Number
Tony Morris	(209) 678-2300

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Phone Number
Morris Jersey Girls	(209) 678-2300

Address	City	State	Zip Code
P.O. Box 580596	Modesto	CA	95358

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code

Street and nearest cross street (if no address)	County

Assessor's Parcel Number	Assessor's Parcel Number County
x017-x043-x023-xxxx	Stanislaus

Last date hauled: 05/16/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,080.00 tons

Manure Solids Content: 50.1 %

Method used to determine amount of manure:

108 Loads @ 10 tons per load

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

nancy Azevedo (Jun 25, 2024 10:48 PDT) — Jun 25, 2024

Operator Signature — Date

nancy Azevedo (Jun 25, 2024 10:48 PDT) — Jun 25, 2024

Hauler Signature — Date

DELLAVALLE™
LABORATORY INC

Tony Morris Dairy
PO Box 580596
Modesto, CA 95358

Account# 00-0026170
Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Tony Morris Dairy

Received: 08/29/2023 15:20
Reported: 08/31/2023 11:31

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H2326-01	DW - North	Ground Water	Sean Locke		08/29/2023 8:30
23H2326-02	DW - South	Ground Water	Sean Locke		08/29/2023 8:35
23H2326-03	DW - Milk Barn	Ground Water	Sean Locke		08/29/2023 8:40
23H2326-04	DW - Heifers	Ground Water	Sean Locke		08/29/2023 8:50

Default Cooler Temperature on Receipt °C: 3.1
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Tony Morris Dairy
PO Box 580596
Modesto, CA 95358

Account# 00-0026170
Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Tony Morris Dairy

Received: 08/29/2023 15:20
Reported: 08/31/2023 11:31

Sample Results

Sample: DW - North
23H2326-01 (Water)

Sampled: 8/29/2023 8:30
Sampled By: Sean Locke

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.18**	mmhos/cm	0.01	1		08/30/23 10:50	SM 2510 B		BEH1339
Electrical Conductivity umhos	**1180**	umhos/cm	10.0	1		08/30/23 10:50	SM 2510 B		BEH1339
Ammonia (as N)	*	mg/L	0.00	1		08/29/23 08:30	Field		BEH1401
Nitrate Nitrogen as NO3N	**31.1**	mg/L	0.1	1	10	08/29/23 22:13	EPA 300.0		BEH1342
pH	**8.0**	units	1.0	1		08/30/23 10:50	SM 4500-H+	H	BEH1339
Temperature	**25.0**	°C	0.0	1		08/30/23 10:50	SM 2510 B		BEH1339
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/31/23 08:33	SM 4500-NH3 C		BEH1420
Total Nitrogen	**31.3**	mg/L	1.00	1		08/31/23 08:33	SM 4500-NH3 C		BEH1420

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DELLAVALLE™
LABORATORY INC

Tony Morris Dairy
PO Box 580596
Modesto, CA 95358

Account# 00-0026170
Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Tony Morris Dairy

Received: 08/29/2023 15:20
Reported: 08/31/2023 11:31

Sample Results
(Continued)

Sample: DW - South
23H2326-02 (Water)

Sampled: 8/29/2023 8:35
Sampled By: Sean Locke

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.58**	mmhos/cm	0.01	1		08/30/23 10:51	SM 2510 B		BEH1339
Electrical Conductivity umhos	**1580**	umhos/cm	10.0	1		08/30/23 10:51	SM 2510 B		BEH1339
Ammonia (as N)	*	mg/L	0.00	1		08/29/23 08:35	Field		BEH1401
Nitrate Nitrogen as NO3N	**51.6**	mg/L	0.1	1	10	08/29/23 22:33	EPA 300.0		BEH1342
pH	**7.6**	units	1.0	1		08/30/23 10:51	SM 4500-H+	H	BEH1339
Temperature	**25.0**	°C	0.0	1		08/30/23 10:51	SM 2510 B		BEH1339
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/31/23 08:34	SM 4500-NH3 C		BEH1420
Total Nitrogen	**51.7**	mg/L	1.00	1		08/31/23 08:34	SM 4500-NH3 C		BEH1420

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Account Manager: Ben Nydam
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Ranch: Tony Morris Dairy

Received: 08/29/2023 15:20
Reported: 08/31/2023 11:31

Sample Results
(Continued)

Sample: DW - Milk Barn
23H2326-03 (Water)

Sampled: 8/29/2023 8:40
Sampled By: Sean Locke

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.55**	mmhos/cm	0.01	1		08/30/23 10:52	SM 2510 B		BEH1339
Electrical Conductivity umhos	**1550**	umhos/cm	10.0	1		08/30/23 10:52	SM 2510 B		BEH1339
Ammonia (as N)	*	mg/L	0.00	1		08/29/23 08:40	Field		BEH1401
Nitrate Nitrogen as NO3N	**50.2**	mg/L	0.1	1	10	08/29/23 22:52	EPA 300.0		BEH1342
pH	**7.5**	units	1.0	1		08/30/23 10:52	SM 4500-H+	H	BEH1339
Temperature	**25.0**	°C	0.0	1		08/30/23 10:52	SM 2510 B		BEH1339
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/31/23 08:36	SM 4500-NH3 C		BEH1420
Total Nitrogen	**50.4**	mg/L	1.00	1		08/31/23 08:36	SM 4500-NH3 C		BEH1420

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Received: 08/29/2023 15:20
Reported: 08/31/2023 11:31

Sample Results
(Continued)

Sample: DW - Heifers
23H2326-04 (Water)

Sampled: 8/29/2023 8:50
Sampled By: Sean Locke

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.99**	mmhos/cm	0.01	1		08/30/23 10:54	SM 2510 B		BEH1339
Electrical Conductivity umhos	**1990**	umhos/cm	10.0	1		08/30/23 10:54	SM 2510 B		BEH1339
Ammonia (as N)	*	mg/L	0.00	1		08/29/23 08:50	Field		BEH1401
Nitrate Nitrogen as NO3N	**31.1**	mg/L	0.1	1	10	08/29/23 23:12	EPA 300.0		BEH1342
pH	**8.0**	units	1.0	1		08/30/23 10:54	SM 4500-H+	H	BEH1339
Temperature	**25.0**	°C	0.0	1		08/30/23 10:54	SM 2510 B		BEH1339
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/31/23 08:37	SM 4500-NH3 C		BEH1420
Total Nitrogen	**31.5**	mg/L	1.00	1		08/31/23 08:37	SM 4500-NH3 C		BEH1420

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Ranch: Tony Morris Dairy

Received: 08/29/2023 15:20
Reported: 08/31/2023 11:31

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1339										
Blank (BEH1339-BLK1)					Prepared & Analyzed: 8/30/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.4		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH1339-BLK2)					Prepared & Analyzed: 8/30/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	6.4		1.0	units						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEH1339-BLK3)					Prepared & Analyzed: 8/30/2023					
pH	6.5		1.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEH1339-DUP1)		**Source: 23H0043-01**			Prepared: 8/29/2023 Analyzed: 8/30/2023					
Electrical Conductivity	2.71		0.01	mmhos/cm		2.78			2.43	10
pH	8.9		1.0	units		8.9			0.113	10
Electrical Conductivity umhos	2710		10.0	umhos/cm		2780			2.43	10
Duplicate (BEH1339-DUP2)		**Source: 23H2332-01**			Prepared: 8/29/2023 Analyzed: 8/30/2023					
pH	7.4		1.0	units		7.4			0.00	10
Electrical Conductivity	0.97		0.01	mmhos/cm		0.97			0.103	10
Electrical Conductivity umhos	969		10.0	umhos/cm		970			0.103	10
Reference (BEH1339-SRM1)					Prepared: 8/29/2023 Analyzed: 8/30/2023					
Electrical Conductivity	584			umhos/cm	538.0		109	90-110		
Reference (BEH1339-SRM2)					Prepared: 8/29/2023 Analyzed: 8/30/2023					
pH	5.8			units	5.820		101	28178-101.7		
Reference (BEH1339-SRM3)					Prepared: 8/29/2023 Analyzed: 8/30/2023					
Electrical Conductivity	990			umhos/cm	1000		99.0	90-110		
Electrical Conductivity umhos	990			umhos/cm	1000		99.0	90-110		
Reference (BEH1339-SRM4)					Prepared: 8/29/2023 Analyzed: 8/30/2023					
Electrical Conductivity	1000			umhos/cm	1000		100	90-110		
Electrical Conductivity umhos	1000			umhos/cm	1000		100	90-110		
Reference (BEH1339-SRM5)					Prepared: 8/29/2023 Analyzed: 8/30/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		

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Received: 08/29/2023 15:20
Reported: 08/31/2023 11:31

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1339 (Continued)										
Reference (BEH1339-SRM5)					Prepared: 8/29/2023	Analyzed: 8/30/2023				
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEH1339-SRM6)					Prepared: 8/29/2023	Analyzed: 8/30/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEH1339-SRM7)					Prepared: 8/29/2023	Analyzed: 8/30/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEH1339-SRM8)					Prepared: 8/29/2023	Analyzed: 8/30/2023				
pH	4.0			units	4.000		101	97.5-102.5		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1342										
Blank (BEH1342-BLK1)					Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH1342-BLK2)					Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH1342-BLK3)					Prepared: 8/29/2023 Analyzed: 8/30/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH1342-BS1)					Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		100	90-110		
LCS (BEH1342-BS2)					Prepared: 8/29/2023 Analyzed: 8/30/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.6	90-110		
Duplicate (BEH1342-DUP1)		**Source: 23H2255-01**			Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	1.7		0.1	mg/L		1.6			1.45	10
Duplicate (BEH1342-DUP2)		**Source: 23H2322-01**			Prepared: 8/29/2023 Analyzed: 8/30/2023					
Nitrate Nitrogen as NO3N	2.4		0.1	mg/L		2.4			0.678	10
Matrix Spike (BEH1342-MS1)		**Source: 23H2255-01**			Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	6.7		0.1	mg/L	5.000	1.6	101	90-110		
Matrix Spike (BEH1342-MS2)		**Source: 23H2322-01**			Prepared: 8/29/2023 Analyzed: 8/30/2023					
Nitrate Nitrogen as NO3N	7.5		0.1	mg/L	5.000	2.4	102	90-110		
Reference (BEH1342-SRM1)					Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		98.8	90-110		
Reference (BEH1342-SRM2)					Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	10.0			mg/L	10.00		99.5	90-110		
Reference (BEH1342-SRM3)					Prepared: 8/29/2023 Analyzed: 8/30/2023					
Nitrate Nitrogen as NO3N	10.0			mg/L	10.00		99.6	90-110		

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LABORATORY INC

Tony Morris Dairy
PO Box 580596
Modesto, CA 95358

Account# 00-0026170
Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Tony Morris Dairy

Received: 08/29/2023 15:20
Reported: 08/31/2023 11:31

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1420										
Blank (BEH1420-BLK1)					Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEH1420-BLK2)					Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEH1420-BS1)					Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	5.63		1.00	mg/L	5.709		98.6	90-110		
LCS (BEH1420-BS2)					Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	5.93		1.00	mg/L	5.709		104	90-110		
Duplicate (BEH1420-DUP1)		**Source: 23H0060-01**			Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	14.9		3.50	mg/L		16.1			7.86	10
Duplicate (BEH1420-DUP2)		**Source: 23H2333-01**			Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Matrix Spike (BEH1420-MS1)		**Source: 23H0060-01**			Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	28.6		3.50	mg/L	9.990	16.1	125	90-110		
Matrix Spike (BEH1420-MS2)		**Source: 23H2333-01**			Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	4.29		1.40	mg/L	7.992	ND	53.7	90-110		
Reference (BEH1420-SRM1)					Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	22.7			mg/L	23.80		95.3	90-110		

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08/29/23 15:20 23H2326

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

No. of **Water** Samples 4 No. of Bottles 4 8

Logged By _____ Results Needed By _____

BILL TO: 260170 / 098 _____
Acct#/Cons. Purchase Order No

TONY MORRIS DAIRY
Company Name

P.O. BOX 580596
Address

MODESTO, CA 95358
City, State Zip

Telephone Fax

Cell/Email

REQUESTED BY F&R Ag Services

PROJECT TONY MORRIS DAIRY

Crop _____ Intended _____

Sampled By SEAN LOCKE

Water Type: [ ] Drinking Water [ ] Wastewater
[ ] Ag Water [X] Groundwater [ ] Monitoring Well
[ ] Supply Water

Other: No. of Samples **Soil** _____ **Tissue** _____ **Manure** _____

Analysis and Bottles Required:

[X] Well Water **(DWW1)** - EC, NO_3-N, Field NH_4-N, if detected run NH_4-N (1-1 liter plastic, unpreserved) **(Annually)** TTU
[ ] Well Water **(DWW2)** - DWW1 plus HCO_3, CO_3, Cl, SO_4, Dissolved: Ca, Mg, Na, TDS (1-1 liter plastic, unpreserved) **(Every 5 yrs)**
[ ] Process Water **(DPW1)** - EC, NO_3-N, NH_4-N, TKN, TDS, TP, TK (1-1 liter plastic, unpreserved) **(Quarterly)**
[ ] Process Water **(DPW2)** - DPW1 plus CO_3, HCO_3, Cl, SO_4, Total: Ca, Mg, Na, (1-1 liter plastic, unpreserved) **(Every 2 yrs)**
[ ] Canal Water **(DCW1)**-EC, NO_3-N,TKN,TN,TDS (1-1 liter plastic,unpreserved)
[ ] Water – EC, NO_3-N, TDS, TP (1-1 liter plastic, unpreserved)
[ ] Manure **(DM1)** - N, P, K, Moisture **(2x/yr)**
[ ] Manure **(DM2)** - DM1 plus Ash, Ca, Mg, Na, S, Cl **(Every 2 yrs)**
[ ] Compost **(OSA2)** – Moisture, N, P, K, Ash, OM, C:N ratio, Ca, Mg, Na, Fe, Cu, Mn, Zn, S
[ ] Harvest Tissue **(CRA1)** - Moisture , N, P, K, Ash
[ ] Tissue – % Moisture
[ ] Fall Soil – **(DS2)** - FA1 (0-1') **plus CEC, OM:** NO_3-N (1-2')
[ ] OTHER:

COPIES TO: labs@fragservices.com, slocke@fragservices.com
[ ] Co Health Dept
[ ] RWQCB [X] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Description of Sample	Field NH_4-N	Date/Time Sampled	Rec'd Temp °C
1. DW- NORTH		8/29 8:30	3.1
2. DW- SOUTH		8:35	3.9
3. DW- MILK BARN		8:40	4.1
4. DW- HEIFERS		8:50	4.2
5			
6.			
7.			
8.			
9.			
10.			

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	[signature]	F+R	8/29 8:50	
Second	[signature]	RLV	8-29-23 2:00	8-29-23 3:08
Third				
Fourth	[signature]	DFI	8/29 1520	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Shipping
$_____ In
$_____ Out

Amt Paid _____ Rec By _____ Check # _____ Date _____

Signature _____

08/29/23 15:20 23H2326

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

☐ Samples refridgerated before pick up | ☐ Picked up samples placed in Ice chest

Container: Ice Chest ☒ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☒ Preserved Upon Receipt at Laboratory

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic	1	1	1	1						
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic	1	1	1	1						
	* pH Value	<2	<2	<2	<2						
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
	40mL AG VOA unpreserved (White) (Set of 3)										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

pH Strips
Lot: 10BDH4501 Exp: Jan 2025

SENT OUT

DELLAVALLE™
LABORATORY INC

Tony Morris Dairy
PO Box 580596
Modesto, CA 95358

Account# 00-0026170
Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Tony Morris Dairy

Received: 08/29/2023 15:20
Reported: 09/01/2023 15:39

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H2331-01	IW #1 Dairy	Ag Water	Sean Locke		08/29/2023 10:00
23H2331-02	IW #2 Carpenter	Ag Water	Sean Locke		08/29/2023 10:15
23H2331-03	IW #3 Vivian	Ag Water	Sean Locke		08/29/2023 10:30

Default Cooler Temperature on Receipt °C: 9.7
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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1910 W. McKinley Ave Suite 110 Fresno, CA 93728 559-233-6129 www.dellavallelab.com

DELLAVALLE™
LABORATORY INC

Tony Morris Dairy
PO Box 580596
Modesto, CA 95358

Account# 00-0026170
Account Manager: Ben Nydam
Submitted By: F & R Ag Services
Ranch: Tony Morris Dairy

Received: 08/29/2023 15:20
Reported: 09/01/2023 15:39

Sample Results

Sample: IW #1 Dairy
23H2331-01 (Water)

Sampled: 8/29/2023 10:00
Sampled By: Sean Locke

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.08**	mmhos/cm	0.01	1		08/30/23 10:56	SM 2510 B		BEH1339
Electrical Conductivity umhos	**1080**	umhos/cm	10.0	1		08/30/23 10:56	SM 2510 B		BEH1339
Nitrate Nitrogen as NO3N	**31.1**	mg/L	0.1	1	10	08/29/23 23:51	EPA 300.0		BEH1342
pH	**7.8**	units	1.0	1		08/30/23 10:56	SM 4500-H+	H	BEH1339
Total Filterable Solids (TDS)	**680**	mg/L	10.0	1		09/01/23 12:59	SM 2540 C		BEH1449
Temperature	**25.0**	°C	0.0	1		08/30/23 10:56	SM 2510 B		BEH1339
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/31/23 08:40	SM 4500-NH3 C		BEH1420
Total Nitrogen	**31.1**	mg/L	1.00	1		08/31/23 08:40	SM 4500-NH3 C		BEH1420

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Received: 08/29/2023 15:20
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Sample Results
(Continued)

Sample: IW #2 Carpenter
23H2331-02 (Water)

Sampled: 8/29/2023 10:15
Sampled By: Sean Locke

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.00**	mmhos/cm	0.01	1		08/30/23 10:57	SM 2510 B		BEH1339
Electrical Conductivity umhos	**997**	umhos/cm	10.0	1		08/30/23 10:57	SM 2510 B		BEH1339
Nitrate Nitrogen as NO3N	**30.0**	mg/L	0.1	1	10	08/30/23 00:11	EPA 300.0		BEH1342
pH	**8.0**	units	1.0	1		08/30/23 10:57	SM 4500-H+	H	BEH1339
Total Filterable Solids (TDS)	**640**	mg/L	10.0	1		09/01/23 12:59	SM 2540 C		BEH1449
Temperature	**25.0**	°C	0.0	1		08/30/23 10:57	SM 2510 B		BEH1339
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/31/23 08:41	SM 4500-NH3 C		BEH1420
Total Nitrogen	**30.6**	mg/L	1.00	1		08/31/23 08:41	SM 4500-NH3 C		BEH1420

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Sample Results
(Continued)

Sample: IW #3 Vivian
23H2331-03 (Water)

Sampled: 8/29/2023 10:30
Sampled By: Sean Locke

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.71**	mmhos/cm	0.01	1		08/30/23 10:59	SM 2510 B		BEH1339
Electrical Conductivity umhos	**1710**	umhos/cm	10.0	1		08/30/23 10:59	SM 2510 B		BEH1339
Nitrate Nitrogen as NO3N	**55.0**	mg/L	0.1	1	10	08/30/23 00:31	EPA 300.0		BEH1342
pH	**7.8**	units	1.0	1		08/30/23 10:59	SM 4500-H+	H	BEH1339
Total Filterable Solids (TDS)	**1100**	mg/L	10.0	1		09/01/23 12:59	SM 2540 C		BEH1449
Temperature	**25.0**	°C	0.0	1		08/30/23 10:59	SM 2510 B		BEH1339
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/31/23 08:42	SM 4500-NH3 C		BEH1420
Total Nitrogen	**55.4**	mg/L	1.00	1		08/31/23 08:42	SM 4500-NH3 C		BEH1420

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Received: 08/29/2023 15:20
Reported: 09/01/2023 15:39

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1339										
Blank (BEH1339-BLK1)					Prepared & Analyzed: 8/30/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.4		1.0	units						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEH1339-BLK2)					Prepared & Analyzed: 8/30/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	6.4		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH1339-BLK3)					Prepared & Analyzed: 8/30/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	6.5		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEH1339-DUP1)		**Source: 23H0043-01**			Prepared: 8/29/2023 Analyzed: 8/30/2023					
Electrical Conductivity	2.71		0.01	mmhos/cm		2.78			2.43	10
pH	8.9		1.0	units		8.9			0.113	10
Electrical Conductivity umhos	2710		10.0	umhos/cm		2780			2.43	10
Duplicate (BEH1339-DUP2)		**Source: 23H2332-01**			Prepared: 8/29/2023 Analyzed: 8/30/2023					
pH	7.4		1.0	units		7.4			0.00	10
Electrical Conductivity	0.97		0.01	mmhos/cm		0.97			0.103	10
Electrical Conductivity umhos	969		10.0	umhos/cm		970			0.103	10
Reference (BEH1339-SRM1)					Prepared: 8/29/2023 Analyzed: 8/30/2023					
Electrical Conductivity	584			umhos/cm	538.0		109	90-110		
Reference (BEH1339-SRM2)					Prepared: 8/29/2023 Analyzed: 8/30/2023					
pH	5.8			units	5.820		101	28178-101.71		
Reference (BEH1339-SRM3)					Prepared: 8/29/2023 Analyzed: 8/30/2023					
Electrical Conductivity	990			umhos/cm	1000		99.0	90-110		
Electrical Conductivity umhos	990			umhos/cm	1000		99.0	90-110		
Reference (BEH1339-SRM4)					Prepared: 8/29/2023 Analyzed: 8/30/2023					
Electrical Conductivity	1000			umhos/cm	1000		100	90-110		
Electrical Conductivity umhos	1000			umhos/cm	1000		100	90-110		
Reference (BEH1339-SRM5)					Prepared: 8/29/2023 Analyzed: 8/30/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1339 (Continued)										
Reference (BEH1339-SRM5)					Prepared: 8/29/2023	Analyzed: 8/30/2023				
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEH1339-SRM6)					Prepared: 8/29/2023	Analyzed: 8/30/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEH1339-SRM7)					Prepared: 8/29/2023	Analyzed: 8/30/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEH1339-SRM8)					Prepared: 8/29/2023	Analyzed: 8/30/2023				
pH	4.0			units	4.000		101	97.5-102.5		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1342										
Blank (BEH1342-BLK1)					Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH1342-BLK2)					Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH1342-BLK3)					Prepared: 8/29/2023 Analyzed: 8/30/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH1342-BS1)					Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		100	90-110		
LCS (BEH1342-BS2)					Prepared: 8/29/2023 Analyzed: 8/30/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.6	90-110		
Duplicate (BEH1342-DUP1)		**Source: 23H2255-01**			Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	1.7		0.1	mg/L		1.6			1.45	10
Duplicate (BEH1342-DUP2)		**Source: 23H2322-01**			Prepared: 8/29/2023 Analyzed: 8/30/2023					
Nitrate Nitrogen as NO3N	2.4		0.1	mg/L		2.4			0.678	10
Matrix Spike (BEH1342-MS1)		**Source: 23H2255-01**			Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	6.7		0.1	mg/L	5.000	1.6	101	90-110		
Matrix Spike (BEH1342-MS2)		**Source: 23H2322-01**			Prepared: 8/29/2023 Analyzed: 8/30/2023					
Nitrate Nitrogen as NO3N	7.5		0.1	mg/L	5.000	2.4	102	90-110		
Reference (BEH1342-SRM1)					Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		98.8	90-110		
Reference (BEH1342-SRM2)					Prepared & Analyzed: 8/29/2023					
Nitrate Nitrogen as NO3N	10.0			mg/L	10.00		99.5	90-110		
Reference (BEH1342-SRM3)					Prepared: 8/29/2023 Analyzed: 8/30/2023					
Nitrate Nitrogen as NO3N	10.0			mg/L	10.00		99.6	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1420										
Blank (BEH1420-BLK1)					Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEH1420-BLK2)					Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEH1420-BS1)					Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	5.63		1.00	mg/L	5.709		98.6	90-110		
LCS (BEH1420-BS2)					Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	5.93		1.00	mg/L	5.709		104	90-110		
Duplicate (BEH1420-DUP1)		**Source: 23H0060-01**			Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	14.9		3.50	mg/L		16.1			7.86	10
Duplicate (BEH1420-DUP2)		**Source: 23H2333-01**			Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Matrix Spike (BEH1420-MS1)		**Source: 23H0060-01**			Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	28.6		3.50	mg/L	9.990	16.1	125	90-110		
Matrix Spike (BEH1420-MS2)		**Source: 23H2333-01**			Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	4.29		1.40	mg/L	7.992	ND	53.7	90-110		
Reference (BEH1420-SRM1)					Prepared: 8/30/2023	Analyzed: 8/31/2023				
Kjeldahl Nitrogen (TKN), Total	22.7			mg/L	23.80		95.3	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH1449										
Blank (BEH1449-BLK1)					Prepared: 8/31/2023	Analyzed: 9/1/2023				
Total Filterable Solids (TDS)	ND		10.0	mg/L						
LCS (BEH1449-BS1)					Prepared: 8/31/2023	Analyzed: 9/1/2023				
Total Filterable Solids (TDS)	17.5		10.0	mg/L	2000		0.875	0-200		
Duplicate (BEH1449-DUP1)			**Source: 23H2331-01**		Prepared: 8/31/2023	Analyzed: 9/1/2023				
Total Filterable Solids (TDS)	690		10.0	mg/L		680			1.46	10
Reference (BEH1449-SRM1)					Prepared: 8/31/2023	Analyzed: 9/1/2023				
Total Filterable Solids (TDS)	323			mg/L	325.0		99.5	90-110		

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08/29/23 15:20 23H2331

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

8-29-23

No. of **Water** Samples 3 No. of Bottles ~~3~~ 6

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well
[ ] Supply Water

Other: No. of Samples **Soil** _____ **Tissue** _____ **Manure** _____

Logged By _____ Results Needed By _____

BILL TO: 26170 / 098 (Acct#/Cons.) Purchase Order No _____

TONY MORRIS DAIRY
Company Name

P.O. BOX 580596
Address

MODESTO, CA 95358
City, State Zip

Telephone _____ Fax _____

Cell/Email _____

REQUESTED BY F&R Ag Services

PROJECT TONY MORRIS DAIRY

Crop _____ Intended _____

Sampled By SEAN LOCKE

Analysis and Bottles Required:

[ ] Well Water **(DWW1)** - EC, NO_3-N, Field NH_4-N, if detected run NH_4-N (1-1 liter plastic, unpreserved) **(Annually)**
[ ] Well Water **(DWW2)** - DWW1 plus HCO_3, CO_3, Cl, SO_4, Dissolved: Ca, Mg, Na, TDS (1-1 liter plastic, unpreserved) **(Every 5 yrs)**
[ ] Process Water **(DPW1)** - EC, NO_3-N, NH_4-N, TKN, TDS, TP, TK (1-1 liter plastic, unpreserved) **(Quarterly)**
[ ] Process Water **(DPW2)** - DPW1 plus CO_3, HCO_3, Cl, SO_4, Total: Ca, Mg, Na, (1-1 liter plastic, unpreserved) **(Every 2 yrs)**
[X] Canal Water **(DCW1)**-EC, NO_3-N,TKN,TN,TDS (1-1 liter plastic,unpreserved)
[ ] Water – EC, NO_3-N, TDS, TP (1-1 liter plastic, unpreserved)
[ ] Manure **(DM1)** - N, P, K, Moisture **(2x/yr)**
[ ] Manure **(DM2)** - DM1 plus Ash, Ca, Mg, Na, S, Cl **(Every 2 yrs)**
[ ] Compost **(OSA2)** – Moisture, N, P, K, Ash, OM, C:N ratio, Ca, Mg, Na, Fe, Cu, Mn, Zn, S
[ ] Harvest Tissue **(CRA1)** - Moisture , N, P, K, Ash
[ ] Tissue – % Moisture
[ ] Fall Soil – **(DS2)** - FA1 (0-1') **plus CEC, OM**; NO_3-N (1-2')
[ ] OTHER:

COPIES TO: labs@fragservices.com, slocke@fragservices.com
[ ] Co Health Dept
[ ] RWQCB [X] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Description of Sample	Field NH_4-N	Date/Time Sampled	Rec'd Temp °C
1. IW #1 DAIRY		8/29 10:00	9.7
2. IW #2 CARPENTER		10:15	7.6
3. IW #3 VIVIAN		10:30	8.6
4.			
5			
6.			
7.			
8.			
9.			
10.			

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	[signature]	F+R	8/29 10:30	
Second	JLF	DLV	8-29-23 2:00	8-29-23 3:08
Third				
Fourth	[signature]	DLV	8/29 15:20	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Shipping
$_____ In
$_____ Out

Amt Paid _____ Rec By _____ Check # _____ Date _____

Signature _____

08/29/23 15:20 23H2331

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

☐ Samples refridgerated before pick up | ☐ Picked up samples placed in Ice chest

Container: Ice Chest ☒ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☒ Preserved Upon Receipt at Laboratory

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic	1	1	1							
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic	1	1	1							
	* pH Value	<2	<2	<2							
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
	40mL AG VOA unpreserved (White) (Set of 3)										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

pH Strips
Lot: 10BDH4501 Exp: Jan 2025

DRAFT
SEND OUT

2023 Tony Morris Annual Report

Final Audit Report 2024-06-25

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